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Where Is AI-Based Crop-Yield Research Done? A Study-Location Analysis of 2,616 Country-Crop Systems

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ABSTRACT

Background: Artificial intelligence (AI) and machine learning now dominate crop-yield prediction, but the geography of that evidence is undocumented: bibliometric studies record where authors work, not where the agriculture studied is located. **Objective:** To measure that distribution across countries and crops and test whether it tracks agricultural scale, research need, or scientific capacity. **Methods:** Screening 43,543 works gave 7,045 eligible studies, 2000–2026. Each was assigned to the countries and crops it empirically examines, read from its reported study area rather than author affiliations. The 2,031 studies with accepted country evidence and a matched crop were allocated fractionally across 2,616 country-crop systems spanning 199 countries and 25 crops. Attention was compared with harvested area, a nine-component research-need index excluding agricultural scale, and capacity indicators, using participation and count regressions with spatial statistics. **Results:** Evidence is narrowly held: 120 of 199 countries carry at least one eligible study, 483 of 2,616 systems carry any research, and the Gini coefficient of country counts is 0.849. The 1,866 unstudied systems with harvested area hold 19.1% of the panel's cropland. Harvested area has the largest association with participation: an interquartile-range increase raises the predicted probability by 35 percentage points, five times the next largest covariate. Attention scales sublinearly with area (elasticity 0.849, unit elasticity rejected, $p = 0.002$). Research and development expenditure predicts participation and intensity, other capacity indicators show no consistent direction, and the research-need association is not identified. Research standing relative to need is spatially clustered (Moran's $I = 0.364$), positive in western and northern Europe, negative in central and eastern Africa. **Conclusions:** AI-based yield evidence follows cropland and research funding, not need: 54 of 188 indexed countries combine above-median need with below-median research. Closing that shortfall calls for data sharing and capacity investment targeted at unstudied systems.

1 | Introduction

Reliable crop-yield information supports decisions made before and during the growing season, including input allocation, production planning, and responses to weather and market uncertainty. At broader scales, yield estimates also inform food-supply assessments, agricultural policy, and resource management [1]. Improving the accuracy and practical value of yield prediction remains an important empirical problem in agricultural research.

Recent reviews document rapid growth in AI- and machine-learning-based crop-yield prediction, with models combining weather, soil, vegetation, and remote-sensing data across a wide range of statistical and machine-learning approaches [2, 1]. The expansion of AI-based yield prediction does not establish that more complex methods uniformly outperform simpler alternatives. Reported accuracy depends on the available data, crop and production setting, and evaluation design [3].

Geographic transferability is therefore central to the interpretation of crop-yield models. High within-sample accuracy does not guarantee similar performance in new fields, regions, or production systems. Studies using spatially independent validation generally report weaker out-of-domain performance than random validation, indicating that spatial structure can inflate apparent predictive accuracy [4, 5, 6]. The geographic scope of the evidence base affects how broadly reported model performance can be interpreted.

The geographic distribution of crop-yield research also reflects a broader pattern in scientific production. Cross-country studies show that research output remains concentrated in countries with stronger scientific institutions and research resources, although participation has broadened over time [7, 8]. Persistent concentration can shape both the problems that receive research attention and the locations where empirical evidence is generated.

The agricultural R&D literature introduces a related distinction among agricultural importance, research need, and scientific capacity. Agricultural importance reflects the scale of crop production, measured here by agricultural scale, primarily harvested area; research need reflects problems such as food insecurity, low productivity, vulnerability, or yield gaps; and scientific capacity reflects the resources and institutions available to conduct research. Those dimensions often diverge geographically [9, 10, 11, 12].

Publication counts alone provide an incomplete measure of research allocation. A country may account for many crop-yield studies because it has a large scientific base, because it is a major producer, or both. Publication totals cannot distinguish research concentration that is proportional to agricultural scale from concentration that reflects unequal scientific capacity. The same problem arises when research attention is compared with research need.

This study addresses that gap by mapping AI- and machine-learning-based crop-yield research using empirical study locations rather than author affiliations. Research attention is evaluated at the country-crop level against production and harvested area, broader indicators of research need, and national scientific capacity. The country-crop design allows research allocation to be compared separately with agricultural scale, research need, and scientific capacity, while also permitting direct assessment of the spatial pattern of resulting mismatches.

2 | Literature Review

2.1 | AI- and Machine-Learning-Based Crop-Yield Prediction

Reviews of AI-based yield prediction converge on the predictors the field relies on. Weather, soil, and vegetation variables recur across studies, and remote sensing now supplies much of the input, with satellite and UAV imagery feeding models that increasingly fuse several data sources rather than depending on one [1, 13, 14, 2].

No single model class dominates the recent literature. Earlier reviews found artificial neural networks the most common approach [1]; later ones report penalised regression, tree ensembles including gradient boosting, support vector machines, and convolutional and recurrent architectures used side by side, together with hybrid and stacked designs [3, 15, 2, 16].

That breadth has produced many reports of strong performance, but not a ranking in which more complex models are reliably better. Reported accuracy varies with data quality, with the crop and production setting, and with how evaluation is carried out [15, 2]. Jabed and Azmi Murad [3] treat yield estimation as a problem shaped by geographical variation, crop diversity, and cultivation area, which makes results from different studies difficult to compare.

The major limitation is comparability across studies rather than the availability of model classes. Reviews report recurring problems with data quality, pre-processing, and model complexity [3, 15], and although RMSE, R^2, and MAE are reported almost everywhere, a shared metric is not a shared validation standard [15, 2]. Growth in this literature is evidence of methodological expansion; it is not yet evidence that performance holds across crops and places.

2.2 | Spatial Transferability and Geographic Representation

Strong performance within the training data does not guarantee reliable prediction elsewhere. Extrapolation across fields and regions is difficult because yield relationships are shaped by spatial autocorrelation, environmental heterogeneity, and processes that operate differently at different scales [4, 6].

Across studies that test this directly, randomly assigned validation folds give a more optimistic view of transferability than geographically independent ones, in deep learning on small fields [6], in soybean prediction from UAV imagery [4], and in NDVI-based winter oilseed rape, where tree ensembles most accurate under random sampling explained almost no variance under field-wise validation [5]. These studies attribute the gap to models memorising spatial structure rather than learning relationships that hold beyond the calibration domain.

Where these studies diverge is on which model classes retain predictive performance under spatial extrapolation, and complexity does not help by default. Habibi, Matsui, and Tanaka [4] found LASSO with recursive feature elimination transferring better than richer models, while Okupska, Juostas, Gozdowski, and Wójcik-Gront [5] found a deep network more stable than tree ensembles under the same spatial split. Validation design determines what a reported accuracy means; the ranking of model families does not survive a change of design. External validity is therefore not a by-product of predictive accuracy, and evidence concentrated in a narrow set of agroecological settings would overstate how broadly its conclusions apply.

2.3 | Scientific Capacity and Agricultural Research Allocation

Research production is itself geographically uneven, and the bibliometric literature is careful about what that unevenness measures. Cash-Gibson, Rojas-Gualdrón, Pericàs, and Benach [8] treat national output as only a crude indication of research capacity, because it also reflects investment, population, and institutional support, and they normalise productivity by population or GDP per capita for that reason.

Concentration is also relational rather than merely distributive: Bueno, Macharete, Rodrigues, Kamia, Moreira, Freitas, Nascimento, and Gadelha [7] find collaboration largely confined within a small core group rather than running between core and periphery, so peripheral systems are not simply smaller but differently connected. Neither study describes a fixed geography, because participation has broadened and capacity outside the traditional core has grown [8].

In the agricultural R&D literature, agricultural importance, research need, and scientific capacity overlap imperfectly: a country may be agriculturally important but have a weak research system, or a strong research system with little dependence on agriculture.

The policy weight attached to public agricultural R&D is visible in the data infrastructure built around it [17]: van Dijk, Fuglie, Heisey, and Deng [12] assemble harmonized public agricultural research spending and personnel for 190 countries from roughly 1960 to

2022, because long-run series are needed to study productivity effects. Rosegrant, Wong, Sulser, Dubosse, and Lybbert [11] estimate a central benefit-cost ratio of 33 for expanded agricultural R&D in the Global South, placing it among the highest-return development investments, while noting that uncertainty in long-run estimation remains a legitimate concern.

Measures of agricultural R&D intensity complicate a simple interpretation of underinvestment. Nin-Pratt [9] argues that the conventional intensity ratio, R&D spending as a share of agricultural GDP, understates the effort developing countries make, yet finds that low- and middle-income countries still face an investment gap close to 50% of R&D spending. Low intensity ratios do not map mechanically onto low effort, but substantial constraints remain. Those constraints track system size and institutional quality: Nin-Pratt and Stads [10] show that small agricultural research systems in low- and middle-income countries tend to have weaker education and research systems, poorer innovation environments, and poorer links to markets, and Stads, Nin-Pratt, Omot, and Thi Pham [18] report Southeast Asian research intensity falling from 0.50% of agricultural GDP in 2000 to 0.33% in 2017 as spending stagnated against rising output.

Research need and agricultural investment are also imperfectly aligned. Zhao and Chen [19] find food-insecure countries receiving only 20% of global agricultural investment, in fewer and smaller projects, and Malec, Rojík, Maitah, Abdu, and Abdullahi [20] show that the effects of agricultural innovation investment on food security vary by subregion within sub-Saharan Africa, so need, impact, and capacity do not align uniformly even within one world region.

2.4 | Existing Agricultural-AI Bibliometrics and Remaining Gap

Bibliometric work on agricultural AI has mapped the field's growth and structure in detail. For crop-yield prediction, these analyses report sustained publication growth and identify the countries, institutions, and themes that dominate it [21, 22]. Momenpour, Bazgeer, and Moghbel [22] additionally note that food security in developing countries receives less attention than its importance would suggest, which is a statement about coverage rather than volume.

The same concentration recurs across broader and more specific bibliometrics: collaboration centred on a handful of countries in agricultural AI [23, 24], most countries contributing fewer than 100 publications each in crop classification from remote sensing [25], and a United States and China core in maize-yield prediction with India and Brazil entering more recently [26]. Satellite-based yield prediction shows the same geographic concentration alongside a shift from regression toward learned models and sensor fusion [27]. The principal geographic unit in these bibliometric studies is author affiliation, institution, or collaboration network rather than the location of the agricultural system examined. Even a crop-specific treatment such as the olive-yield literature is organised around co-authorship structure and citation performance rather than around the production systems the studies examine [28] (Table 1).

The literature identified in this review does not allocate empirical studies to the country-crop systems they examine while jointly comparing research allocation with agricultural scale, research need, and scientific capacity. Doing so requires coding each study's empirical location rather than its authors' affiliations, and evaluating the result against agricultural and socioeconomic denominators at the same unit.

3 | Data

The analysis draws on four bodies of data: a frozen set of eligible AI/ML crop-yield studies coded to the country-crop systems each study examines, FAOSTAT production and harvested-area series, country-level indicators of research need and scientific capacity, and a country boundary layer. Table 2 gives the source, unit, reference period, and analytical role of each. Reproducibility detail supporting this section is in Appendices Appendix A–Appendix C.

3.1 | AI/ML Crop-Yield Study Database

The study database was built from OpenAlex. Retrieval returned 102,892 records, reduced to 43,543 unique works after duplicate resolution. All 43,543 were screened on title and abstract against six eligibility criteria. An eligible study reports an empirical application in which the yield of an agricultural crop is the modelled outcome, estimated with a data-driven predictive method, in an eligible document type published from 2000 onward; process-based crop simulation without a learned component does not qualify. Screening left a frozen corpus of 7,045 eligible studies published between 2000 and 2026. Appendix A reports the retrieval design, the duplicate-resolution rules, the screening procedure, and the measured screening error rates.

3.2 | Study-Location and Country-Crop Coding

The unit of analysis is the country-crop system, so each study is assigned to the countries and crops it empirically examines. Study location is coded from what a study reports about the origin of its agricultural data, never from author affiliation. The two differ systematically, and coding from affiliation would reproduce the geography of authorship rather than measure the geography of evidence. Country names are accepted only where the surrounding text asserts an empirical study area; names appearing in institutional, bibliographic, or varietal contexts are rejected. Studies that are global or regional in scope, that name no study area, or that name only an ambiguous place receive distinct scope labels. Where the coding accepted explicit country evidence inside such a study, those countries enter the allocation; a study with no accepted country evidence stays unallocated, so absence of a coded location remains separable from absence of research.

An automated, model-assisted extractor produced the location and crop coding rather than by human reading, and every record is stored with its evidence text, a confidence level, and an unverified status. We estimated accuracy from a stratified probability sample of 210 records adjudicated against the coded evidence. The countries that receive research weight are correct for 95.6% of adjudicated records (95% CI 90.7 to 98.0), and 0.7% are allocated wholly to the wrong country. Recall is the weaker property: 62.0% of sampled unresolved records (95% CI 48.2 to 74.1) name a study area that the coder did not resolve. Appendix B reports the design, the error taxonomy, and a check on whether the unresolved studies differ geographically.

Crops are standardised to a controlled vocabulary and matched to FAOSTAT items. Studies covering several countries or crops are allocated fractionally, each carrying a total weight of one divided equally across the combinations it examines, which preserves the study total and prevents multi-country work from counting repeatedly. Full counts are retained for comparison. Accepted country evidence and

TABLE 1 | How this study differs from related strands of literature

Literature strand	Primary research focus	Typical geographic unit	Typical outcome	Main limitation relative to the present study
AI/ML crop-yield prediction	Predictor sets, model families, and reported accuracy	Study site, field, or administrative region	Predictive performance for a fitted crop and place	Site selection is not the object of study, so coverage across crops and countries is incidental
Spatial transferability and external validation	Whether fitted relationships hold outside the calibration domain	Fields or spatial folds within one study area	Performance loss under spatially aware validation	Establishes that geography matters for validity without mapping where the evidence base sits
Scientific capacity and agricultural research allocation	Research investment, capacity, and returns	Country, region, or income group	R&D spending, intensity ratios, and productivity or welfare returns	Measures money and capacity rather than published evidence, and rarely reaches the crop level
Agricultural-AI and crop-yield bibliometrics	Publication growth, contributors, collaboration, and themes	Author affiliation country or institution	Output counts, co-authorship networks, and keyword structure	Records where work is authored rather than where the agriculture studied is located, and uses no agricultural denominator
Present study	Global distribution of AI/ML crop-yield evidence relative to agricultural scale, research need, scientific capacity, and spatial concentration	Country-crop study-location unit	Research allocation and mismatch relative to production, harvested area, need, and capacity	—

a FAOSTAT-matched crop coincide for 2,031 studies, which are the studies that enter the allocation. Appendix B gives the coding rules, the cue taxonomy, the resolution counts, and the full accounting from corpus to panel.

3.3 | Agricultural Scale

Agricultural scale is the empirical proxy for agricultural importance, measured primarily by harvested area and secondarily by production compared within a crop. Harvested area does not measure economic value, food-security significance, or strategic priority, and no claim of that kind is drawn from it. FAOSTAT production and harvested-area data provide the benchmarks. Each country-crop cell takes the mean of the 2018–2022 reference period, a window balancing currency against year-to-year variation in weather and reporting. Availability of production and harvested-area values determines which cells can enter denominator-based comparisons. Appendix C reports the extract, the identifier crosswalk, and the resulting cell counts.

3.4 | Research Need and Scientific Capacity

We measure need and capacity separately so that the two can be tested against each other. The research-need index combines nine country-level indicators spanning food security, agricultural dependence, climate exposure, and production instability, each converted to a percentile rank before aggregation. Two further FAOSTAT quantities, a country's share of global harvested area and of global production, are recorded with the need data but excluded from the index by design: they are the paper's agricultural-importance measures, and an index containing them could not support a claim that importance and need are measured separately. Rank aggregation is the primary index; equal weighting, entropy weighting, the first principal component, and an eleven-component variant that adds the two scale shares back are retained for sensitivity analysis.

Scientific capacity uses World Bank indicators of research and development effort, economic resources, tertiary education, digital access, and population. Research and development expenditure as a share of GDP is the primary capacity measure and the sparsest series, so its coverage governs the estimation sample; GDP per capita, tertiary enrolment, internet use, and population enter alongside it. Missing capacity values are not imputed. Appendix C and Table 3 give the full indicator list, sources, coverage, directions, transformations, and missingness.

3.5 | Spatial Data

Country geometries come from the Natural Earth admin-0 boundaries, giving an analytical layer of 195 countries. Four panel territories have no matching geometry and cannot enter the spatial analysis; all four carry no eligible study, so their absence removes no research observation.

3.6 | Analytical Sample

The master panel is the set of country-crop cells for which FAOSTAT reports the crop in the country at any year from 1990 to 2024, taken over the 199 countries and 25 crops carried through the crosswalk, together with 30 cells for four territories that FAOSTAT reports inside another country. That rule gives 2,616 cells, of which 485 carry at least one eligible study and 2,131 carry none. A further 109 cells carry no FAOSTAT area, production, or yield value for the reference period and support no denominator-based comparison. None of the 109 carries research, and restricting the panel to the 2,507 cells with a FAOSTAT observation raises the share of studied systems from 18.5% to 19.3%. The 2,131 empty cells record zero coded research attention: the search covered every country, so a zero means no resolved eligible study was allocated to that cell, not that no eligible study exists for it. Zeros stay distinct from cells where external data are unavailable. Analytical components use different subsets, and the differences are large enough that one sample size would misrepresent them. Comparisons against agricultural scale use cells with a positive production or harvested-area denominator; spatial analysis uses the 195 countries in the boundary layer; the econometric models use the 1,799 cells across 142 countries for which every covariate is observed, research and development expenditure being the covariate with the most missing data. Per-covariate missingness is in Table 3. No country is excluded on grounds of research output. Table 5 summarises the variables entering the models, and Figure 2 traces the path from retrieved records to the estimation sample.

4 | Methods and Procedures

The analysis proceeds in four steps: eligible studies are allocated to country-crop systems to measure research attention; attention is compared with agricultural scale and with research need; the resulting country-level mismatch is tested for spatial dependence; and regression models ask whether attention tracks need or scientific capacity once agricultural size is held fixed. Table 4 maps each research question to the measure and procedure used.

4.1 | Measuring Research Attention and Research Intensity

The empirical unit is the country-crop cell. Let i index countries and k crops, and let s index eligible studies. A study examining n_s^C countries and n_s^K crops is divided equally across the combinations it covers, so that research attention in cell (i, k) is

$$R_{ik} = \sum_{s \in S_{ik}} w_s, \qquad w_s = \frac{1}{n_s^C\, n_s^K}, \tag{1}$$

where S_{ik} is the set of studies covering cell (i, k). A cell with no eligible study takes $R_{ik} = 0$; full counts are retained for sensitivity analysis.

Research intensity normalises attention by the size of the agricultural activity at stake. Writing A_{ik} for mean harvested area and P_{ik} for mean production over the reference period,

$$I_{ik}^A = \frac{R_{ik}}{A_{ik}/10^6}, \qquad I_{ik}^P = \frac{R_{ik}}{P_{ik}/10^6}, \tag{2}$$

each defined only where the denominator is strictly positive, so higher values indicate more research per unit of cultivated area or output. A complementary comparison places attention and production on the same scale as shares of their global totals,

$$\Delta_{ik} = \frac{R_{ik}}{\sum_{jl} R_{jl}} - \frac{P_{ik}}{\sum_{jl} P_{jl}}, \tag{3}$$

so that $\Delta_{ik} > 0$ marks a cell holding a larger share of the literature than of world production. Both terms are shares of the global total across all cells, not shares within a crop.

4.2 | Measuring Research Alignment and Mismatch

We assess alignment against two distinct benchmarks. Equations (2) and (3) compare attention with agricultural scale. The second comparison, against research need, requires a country-level measure because the need index is defined for countries rather than for country-crop systems. Cell-level attention is aggregated, $R_i = \sum_k R_{ik}$, which preserves the fractional weights exactly.

Countries are then ranked on attention and on need, and the two ranks are differenced. Let ρ_i^R denote a country's percentile rank on R_i and ρ_i^N its percentile rank on the research-need index, both expressed on a 0–100 scale. The mismatch measure is

$$G_i = \rho_i^R - \rho_i^N, \tag{4}$$

measured in percentile points. Positive values mark countries better represented in the literature than their need rank implies, negative values the reverse, and zero equal standing in the two distributions. The measure is defined only for countries with a computable need index and is missing rather than imputed elsewhere. No composite combines importance, need, and capacity: each enters separately.

4.3 | Research-Need Measure

The nine components are harmonised in direction so that a higher value always indicates greater need, with dietary energy adequacy and agricultural value added per worker reversed accordingly, then converted to percentile ranks across countries.

Rank aggregation is the primary index: the mean of a country's observed component percentile ranks, re-expressed as a percentile on a 0–1 scale. A country is indexed when it observes at least five of the nine components (188 of 199 countries). The floor counts the nine need components only. Counting the two agricultural-scale shares toward it, as an earlier version of the index did, would admit five further countries on as few as one observed need component, because those two shares are never missing; that variant is reported for comparison in Appendix C. Equal weighting, entropy weighting, the first principal component, and an eleven-component variant are built for sensitivity analysis, though rank aggregation and entropy weighting proved almost indistinguishable, so entropy weighting is not read as independent corroboration. Restricting the index to countries with at least seven or nine observed components leaves the results in Section 5 unchanged. Equation (4) takes the index rescaled to 0–100; the models use the 0–1 scale, so a coefficient reads as the change between the least and the most needy country. Yield volatility enters research need through the index and appears nowhere else in the models.

4.4 | Empirical Framework: Research Need versus Scientific Capacity

Whether a system is studied at all is binary; how much it is studied is a non-negative count with a large mass at zero. We estimate participation and intensity separately.

TABLE 2 | Data sources and measures used in the analysis

Data component	Main measures	Source	Unit	Reference period	Analytical role
AI/ML crop-yield studies	Count of eligible studies, fractional and full	OpenAlex, screened corpus	Study	2000–2026	Research-side outcome; 7,045 studies
Study-location and crop coding	Empirical study country; standardised crop	Study text, coded	Country-crop	2000–2026	Allocates 2,031 studies to the panel
Agricultural scale	Production (t); harvested area (ha)	FAOSTAT crop production	Country-crop	2018–2022 mean	Denominator for research intensity and share comparisons
Research need	Nine indicators of food security, agricultural dependence, climate exposure, and production instability; agricultural-scale shares recorded but excluded from the index	FAOSTAT food security, macro, employment, and temperature domains	Country	Latest available	Rank-aggregation index; four alternatives for sensitivity
Scientific capacity	R&D expenditure; GDP per capita; tertiary enrolment; internet use; population	World Bank indicators	Country	Latest available	Model covariates; R&D coverage governs the estimation sample
Spatial data	Country polygons and centroids	Natural Earth admin-0	Country	Static	195 countries; basis for spatial weights

Reference periods refer to the observations entering the analysis rather than to the full extent of each source.

We model participation as

$$\Pr(y_{ik} = 1 \mid \mathbf{x}_{ik}) \;=\; \Lambda\big(\beta_0 + \mathbf{x}_{ik}'\boldsymbol{\beta}\big), \qquad y_{ik} = \mathbf{1}\{R_{ik} > 0\}, \tag{5}$$

where a probit link is estimated on the same sample as an alternative. The covariate vector $\mathbf{x}_{ik}$ contains log harvested area, the research-need index, research and development expenditure as a share of GDP, log GDP per capita, tertiary enrolment, internet use, and log population. Log harvested area captures agricultural scale, the need index captures need, and the remaining five are capacity-related indicators. Yield volatility enters only through the need index. Crop fixed effects are added in a robustness specification, so the harvested-area association is also estimated from variation across countries within the same crop.

We estimate intensity by Poisson pseudo-maximum likelihood,

$$\mathbb{E}[R_{ik} \mid \mathbf{x}_{ik}] \;=\; \exp\big(\beta_0 + \mathbf{x}_{ik}'\boldsymbol{\beta} + \ln A_{ik}\big), \tag{6}$$

where log harvested area enters as an offset with unit coefficient, so the model describes studies per hectare and area leaves the covariate vector. Poisson pseudo-maximum likelihood accommodates the fractional counts [29]. A The unit elasticity that the offset imposes is tested rather than assumed: a specification estimating the harvested-area coefficient freely is reported alongside, with a Wald test of $\beta_{\text{area}} = 1$. A negative binomial model tests whether overdispersion argues against the Poisson mean assumption, and a quasi-Poisson variance assumption is reported as a further comparison.

Standard errors are clustered on country throughout, because cells within a country share capacity covariates and are not independent draws. Dropping systems missing any covariate leaves 1,799 cells across 142 countries. Coefficients describe association: capacity, need, and agricultural scale are jointly determined with research output, no instrument is used, and no causal interpretation is offered.

4.5 | Spatial Analysis of Research Mismatch

We evaluate spatial dependence on the country-level mismatch measure G_i from (4), and on country aggregates of research output for comparison. A row-standardised k-nearest-neighbour matrix with $k = 6$ is primary because queen contiguity strands 43 islands. Neighbours are found among polygon centroids projected to Equal Earth; results are unchanged when recomputed on great-circle distance, which changes the neighbour set for 86 of 195 countries. Four alternatives are carried through the robustness suite, and the matrices are also compared on a common connected subsample so that weight-matrix sensitivity is not confounded with sample composition (Appendix D).

Global spatial dependence is measured by Moran's I [30],

$$I \;=\; \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}\,(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}, \tag{7}$$

Inference uses 9,999 conditional permutations. Countries missing the variable under test are excluded and the weight matrix is rebuilt on the observed sample, so no filled value enters a reported statistic. Local Moran statistics [31] classify each country by whether a high or low value sits among high or low neighbours, giving High–High, Low–Low, High–Low, and Low–High categories. The p-values are adjusted by the Benjamini–Hochberg procedure [32] at $\alpha = 0.05$ and a country is called a cluster only on the adjusted value. Getis–Ord G_i^* [33] is computed alongside under the same scheme. Local statistics use the primary matrix only. Residual spatial dependence is tested by applying (7) to country-mean residuals from both models.

TABLE 3 | Variable definitions, coverage, transformations, and missingness

Variable	Source	Period	Direction	Country coverage	Transformation and missingness
Research-need components					
Undernourishment (%)	FAOSTAT food security	2001–2024	Higher	172	Percentile rank
Food insecurity, moderate or severe (%)	FAOSTAT food security	2015–2024	Higher	197	Percentile rank
Cereal import dependency (%)	FAOSTAT food security	2001–2022	Higher	220	Percentile rank
Dietary energy adequacy (%)	FAOSTAT food security	2001–2024	Lower	202	Percentile rank
Employment in agriculture (% of employment)	FAOSTAT employment	1991–2025	Higher	226	Percentile rank
Agricultural value added (% of GDP)	FAOSTAT macro	1990–2024	Higher	244	Percentile rank
Agricultural value added per worker (USD)	FAOSTAT employment	1991–2024	Lower	183	Percentile rank
Temperature change since the 1951–1980 baseline (°C)	FAOSTAT temperature	1990–2025	Higher	285 in source, 190 in index	Percentile rank
Area-weighted yield volatility	Derived, FAOSTAT crops	2018–2022	Higher	194	Percentile rank
Share of global harvested area	Derived, FAOSTAT crops	2018–2022	—	199	Recorded with the need data; excluded from the index (importance measure)
Share of global production	Derived, FAOSTAT crops	2018–2022	—	199	Recorded with the need data; excluded from the index (importance measure)
Scientific-capacity indicators, with missing panel cells					
R&D expenditure (% of GDP)	World Bank	from 1996	—	192	Latest observation; 514 cells missing
GDP per capita, PPP	World Bank	from 1990	—	242	Logged; 154 cells missing
Tertiary enrolment (%)	World Bank	from 1990	—	241	125 cells missing
Internet users (%)	World Bank	from 1990	—	241	56 cells missing
Population, total	World Bank	from 1990	—	260	Logged; 56 cells missing; paging repair applied
Employment in agriculture (%)	World Bank	from 1991	—	230	Retained; not in final specification
Other panel quantities					
Harvested area	FAOSTAT crops	2018–2022	—	199	Logged; 112 cells missing
Yield volatility (CV)	Derived, FAOSTAT crops	2018–2022	—	199	248 cells missing; enters models only through the need index

Direction indicates whether a higher value corresponds to greater research need. Country coverage is the number of countries with at least one observation in the source extract. Missing cells are counts within the 2,616-cell panel for covariates entering the econometric models.

4.6 | Robustness and Sensitivity Analysis

Fifteen variants test whether measured concentration depends on counting rules, document type, location-coding confidence, the allocation scope, the method definition, the largest contributing countries, or the period examined. Fourteen re-estimate the participation model across country exclusions, income groups, crop and region fixed effects, and alternative need indices. Spatial results are re-computed across neighbour definitions, a common connected subsample, and both need indices. Results are reported rather than graded. For each quantity Table 6 gives the baseline estimate, the range across variants, and how many variants retain the sign and exclude zero. Thresholds separating stable from unstable findings were not fixed before the results were examined, so no categorical label is attached to them and the spread is given instead. Quantities whose intervals contain zero are reported as such rather than omitted. Appendix D lists the specifications.

5 | Results

5.1 | Global Distribution of AI/ML Crop-Yield Research

Research attention concentrates in a small number of countries and crops. Of the 199 countries in the panel, 120 carry at least one eligible study and 79 carry none, and only 483 of the 2,616 country-crop systems contain any research at all (Figure 1). The Gini coefficient of fractional study counts across all 199 countries is 0.849, the principal concentration measure because the 79 countries with no resolved study are part of the distribution being described. Conditioning on the 120 countries that carry a study gives 0.749, reported for comparison.

The distribution is top-heavy at every level. The most-studied country accounts for 16.2% of fractional output, the top three for 43.1%, and the top twenty for 73.7%. The United States, India, and China lead with 327.7, 292.6, and 252.3 fractional studies. By income group, upper-middle-income countries hold 34.0% of fractional output and high-income countries 33.7%, against 4.2% for low-income countries.

TABLE 4 | Empirical measures and analytical procedures

Question	Measure	Unit	Procedure
Where is research done?	Fractional study count, Eq. (1)	Country-crop	Allocation and description
Is attention proportional to agriculture?	Intensity and share difference, Eqs. (2)–(3)	Country-crop	Normalisation against harvested area and production
Is attention aligned with need?	Mismatch measure, Eq. (4)	Country	Percentile ranking and differencing
Need or capacity?	Participation and intensity, Eqs. (5)–(6)	Country-crop	Logit and probit; Poisson pseudo-maximum likelihood with area offset; clustered by country
Is mismatch clustered?	Moran's I and local statistics, Eq. (7)	Country	Permutation inference with Benjamini–Hochberg adjustment

TABLE 5 | Summary statistics for the variables entering the regression models

Variable	N	Mean	SD	Median	Min	Max	Mean, $y = 1$	Mean, $y = 0$
Log harvested area	1799	10.215	3.048	10.499	0.000	17.621	12.884	9.333
R&D expenditure (% GDP)	1799	0.806	1.051	0.392	0.006	6.346	1.215	0.671
Research-need index (0-1)	1799	0.513	0.283	0.523	0.016	1.000	0.399	0.551
Log GDP per capita	1799	9.599	1.087	9.726	6.972	11.786	9.941	9.486
Tertiary enrolment (share)	1799	0.467	0.326	0.455	0.019	1.651	0.579	0.430
Internet use (share)	1799	0.705	0.261	0.820	0.086	1.000	0.775	0.682
Log population	1799	16.835	1.632	16.903	11.512	21.104	17.721	16.543
Fractional study count	1799	1.084	6.641	0.000	0.000	159.519	4.361	0.000

The estimation sample is 1,799 country-crop cells across 142 countries. $y = 1$ marks cells carrying at least one eligible study (447 cells); $y = 0$ marks cells carrying none (1,352 cells).

Crop coverage is similarly narrow. Wheat, maize, and rice together take 69.5% of fractional crop attention (565.6, 450.3, and 389.8 fractional studies), and the Gini across crops is 0.721 (Figure A1, Appendix E). The literature is also recent: the median publication year is 2023 and 80.8% of eligible studies appeared in 2018 or later.

5.2 | Research Attention Relative to Agricultural Scale

Research attention tracks agricultural scale only loosely at the country-crop level. Harvested area is the cross-crop benchmark, because a hectare is comparable across crops while a tonne of sugarcane and a tonne of wheat are not; production is compared only within a crop. The Spearman correlation between fractional study counts and harvested area is 0.523 across systems, rising to 0.743 when both are aggregated to countries (0.518 and 0.782 for production), so the correspondence that appears at country level weakens once the crop dimension is restored (Figure 3). The gap between the two correlations is itself the finding: national research effort tracks total cropland, but within a country the distribution across crops does not match the land each occupies. A country can look well covered while most of its cropping systems are unstudied. Attention also scales less than proportionally with area: regressing log research share on log area share across the 477 systems positive in both gives a slope of 0.346 (95% CI 0.258 to 0.434), rejecting proportionality ($\chi^2_1 = 211.1$, $p < 0.001$).

Departures from proportionality run in both directions. Against harvested area, the three major cereals absorb more attention than the land they occupy: wheat holds 28.0% of fractional attention on 17.9% of panel area, maize 22.3% on 16.5%, and rice 19.3% on 13.6%. Oil palm is the clearest shortfall: 0.4% of attention on 2.2% of area, and Indonesia, with 48.5% of world oil-palm area, carries no eligible oil-palm study at all. Within single crops, production-based comparisons point the same way: Brazil holds 24.5% of world sugarcane research against 38.7% of world sugarcane production, and China 12.2% of rice research against 27.3% of rice production, while United States maize and Chinese wheat run above their within-crop production shares. Comparisons that pool production tonnage across crops overstate the shortfall for high-biomass crops; sugarcane holds a quarter of panel tonnage but 2.2% of panel area, and against area its attention share of 3.9% is no longer a deficit.

5.3 | Research Need and Scientific Capacity

Research attention differs from the geographic ranking of research need. Ranking countries on fractional study counts and on the rank-aggregation need index and differencing the two percentiles gives a mismatch measure defined for 188 countries, with 54 falling in the high-need, low-research quadrant (Figure 4); the eleven-component variant that adds the agricultural-scale shares back gives 48, because large producers then count as needier. The countries with the largest shortfalls are concentrated among low-income economies, and 68 of the 188 tie at the lowest research percentile because none carries an eligible study. The regression cannot distinguish among those zeros: a third of the world's countries are tied at zero research, so their need comparison rests on absence of evidence rather than quantity.

Attention is instead concentrated among higher-capacity countries. The 4.2% of fractional output held by low-income countries sits against their far larger share of the countries in the panel, and the same countries occupy the lower end of the research and development expenditure distribution. Whether that association survives controls for agricultural scale is the question the regressions address.

5.4 | Regression Results

Agricultural scale and several capacity indicators are associated with where research is conducted, whereas the association with measured research need is imprecisely estimated. Table 7 reports the participation logit and the intensity model, estimated on 1,799 and 1,792 country-crop systems across 142 countries, with standard errors clustered by country.

Harvested area has the largest and most precisely estimated association with whether a country-crop system is studied at all, with a logit coefficient of 0.723 (95% CI 0.622 to 0.824). The comparison across covariates does not rest on raw coefficients: an interquartile-range increase in log harvested area raises the predicted probability of participation by 35 percentage points, about five times the corresponding change for any other covariate, the largest of which is 7 points for log GDP per capita. Average marginal effects agree: harvested area at 0.081 ($p < 0.001$) and research and development expenditure at 0.036 ($p = 0.010$) are the only covariates distinguishable from zero (Table 8). Figure 6 plots the fitted participation probability across the observed range of harvested area. Adding crop fixed effects, so that the association is identified from variation across countries within the same crop, moves the coefficient to 0.668, and estimating the model separately for each of the five most studied crops returns coefficients from 0.741 to 1.168, all with intervals excluding zero (Table A3). Research and development expenditure is also associated with participation, at 0.323 (CI 0.081 to 0.565), and with intensity once harvested area is held fixed as an exposure offset, at 0.521 (CI 0.281 to 0.760); an interquartile-range increase multiplies expected intensity by 1.5. Internet use and tertiary enrolment are distinguishable from zero only in the intensity model, internet use positively (2.167, CI 0.537 to 3.797) and tertiary enrolment negatively (−0.961, CI −1.798 to −0.124). Log GDP per capita and log population are not distinguishable from zero in either model. R&D expenditure is associated with both participation and intensity, whereas internet use and tertiary enrolment are distinguishable from zero only in the intensity specification.

The need association is not distinguishable from zero once the intensity model drops a restriction the data reject. In the participation model the need coefficient is 0.880 with a standard error of 0.957 and a 95% interval from −0.997 to 2.756: positive but imprecisely estimated, an interval wide enough to contain a large association in either direction. In the intensity model with harvested area entered as an exposure offset it is 1.282 with an interval from 0.139 to 2.424, and it stays marginally distinguishable from zero with crop fixed effects (1.248, CI 0.068 to 2.428). That specification fixes the harvested-area elasticity at one. Estimated freely, the elasticity is 0.849 (SE 0.048, CI 0.755 to 0.944), and a Wald test rejects unit elasticity ($\chi^2_1 = 9.73$, $p = 0.002$); under the free specification the need coefficient falls to 0.982 with an interval from −0.148 to 2.111. Once that restriction is dropped, no specification distinguishes the need association from zero. That is a statement about the precision these data support, not evidence that no association exists. The elasticity itself carries a substantive reading that the offset conceals. At 0.849 rather than one, research per hectare falls as systems grow: doubling harvested area raises expected attention by about 80% rather than 100%, so the largest systems are studied less intensively than their size alone implies, even though they are studied most in absolute terms. Index construction affects the size of the need estimate without changing the conclusion: under the eleven-component variant the two need coefficients are 0.001 and 0.114, against 0.880 and 1.282 under the scale-excluded measure. Figure A2 in Appendix E plots every estimate with its interval.

The participation model achieves a McFadden pseudo-R^2 of 0.365 on 446 events. The intensity model retains 1,792 of the 1,799 cells because the exposure offset $\ln A_{ik}$ is undefined for seven covariate-complete cells recording zero harvested area. The counts are overdispersed, with a conditional variance about forty times the mean (overdispersion ratio ≈ 40). Alternative variance assumptions widen the intervals without changing the point estimates, and a negative binomial returned no standard errors under the clustered covariance estimator; the dispersion diagnostics are reported in Table A7. Neither model has significant residual spatial autocorrelation. Moran's I on country-mean residuals is −0.029 ($p = 0.324$) for participation and −0.020 ($p = 0.386$) for intensity, and Lagrange Multiplier tests reject neither a spatial lag nor a spatial error process for either model under the primary weights (Table A7).

5.5 | Spatial Clustering of Research Mismatch

The mismatch between research standing and need is spatially clustered. Global Moran's I on the country-level mismatch measure, computed over the 188 countries with an observed value, is 0.364 ($p = 0.0001$, 9,999 permutations) under the primary six-nearest-neighbour matrix. The statistic moves little with the neighbour definition, from 0.353 to 0.387 across the k-nearest-neighbour alternatives, and rebuilding the neighbour sets on great-circle rather than projected distance leaves it near the same value. The measure also survives the tie convention: 68 of the 188 countries tie at the lowest research percentile, and the statistic runs from 0.298 to 0.376 across four rank conventions and reaches 0.190 under a standardised log-count alternative that preserves zeros, with $p = 0.0001$ throughout. Counting the two agricultural-scale shares toward the coverage floor, which admits five thinly measured countries, gives 0.366, and the eleven-component index gives 0.398. Mismatch is the most strongly clustered of the quantities examined: the log fractional study count returns 0.157 ($p = 0.0007$) and crop-system coverage 0.143 ($p = 0.0006$), while the citation-weighted count returns 0.022 ($p = 0.088$) and is not significant under the primary matrix.

Local statistics identify clustering in both directions. After Benjamini–Hochberg adjustment, 28 countries carry a significant local statistic: 18 form High–High clusters, 6 Low–Low, 2 High–Low, and 2 Low–High (Figure 5). Seventeen of the 18 High–High countries lie in western and northern Europe, the exception being Brunei Darussalam. The Low–Low cluster holds the Republic of the Congo, the Democratic Republic of the Congo, Gabon, Sudan, Djibouti, and Solomon Islands; Ethiopia is a High–Low outlier, better represented than its low-mismatch neighbours. Getis–Ord G_i^* agrees in direction, returning 20 hot spots and 8 cold spots. The High–High countries form a contiguous bloc, and the Low–Low countries are likewise adjacent to one another, so research shortfalls are spatially concentrated rather than isolated. The eleven-component index gives the same geography with a broader African Low–Low set (19 High–High, 11 Low–Low).

Research output on its own clusters differently. Its local statistics after adjustment give 21 Low–Low clusters against 2 High–High, and Getis–Ord returns 3 hot spots against 21 cold spots, so raw output clusters as shared absence while the need-adjusted mismatch clusters in both directions.

5.6 | Robustness and Sensitivity Results

Table 6 gives each estimated quantity, its baseline, and its spread across the robustness suite. Concentration is stable across publication periods (Figure A3). Splitting the outcome by publication period leaves the area association unchanged, at 0.750 for studies published before 2020 and 0.748 for those published from 2020 onward, while the research and development coefficient falls from 0.567 to 0.285 and the need coefficient stays imprecise in both periods (Table A8).

Concentration is the most durable result. Across the 14 variants that recompute it, the all-country Gini stays between 0.772 and 0.899 against a baseline of 0.847, and removing the United States, China, and India together still leaves 0.772. Journal-only estimation returns 0.849 on 1,712 studies, so conference proceedings do not drive the pattern, and excluding cells whose location rests on the weaker locative cue returns 0.849. Restricting allocation to studies whose overall scope names specific countries gives 0.839, and restricting the corpus to studies whose title or abstract carries an explicit machine-learning method term gives 0.870, so neither the allocation rule nor the breadth of the method definition drives the concentration. The conditional Gini over countries that carry research moves between 0.619 and 0.836 across the same variants.

The area result holds as widely: across 14 model variants, including crop and region fixed effects and both need indices, the logit coefficient on log harvested area stays between 0.654 and 0.861 and is positive with an interval excluding zero in every one. The research and development coefficient is less settled, ranging from −0.074 to 2.218 and moving materially across income-group splits, so it excludes zero in 8 of the 14 variants. Clustering of the mismatch measure holds across neighbour definitions, index variants, great-circle distance, and tie conventions. Clustering of the citation-weighted count ranges from 0.014 to 0.147 and excludes zero in only 1 of 5 variants, so no clustering claim is made for it. The need association with participation has a standard error near one in every variant and excludes zero in none; the need association with intensity is positive under the primary index with and without crop fixed effects, but indistinguishable from zero under the eleven-component variant and under the free-elasticity specification. Adding the capacity indicators to a model already containing agricultural scale and research need improves fit, whereas adding research need to scale and capacity does not (Table A4). Aggregating to countries reproduces the pattern only in part (Table A5). A country-level Poisson model with an area offset keeps the research and development association positive (0.278, CI 0.069 to 0.487) and leaves need imprecise (0.533, CI −0.638 to 1.704). A log-count specification instead returns a negative need coefficient (−1.083, CI −2.072 to −0.094), an imprecise research and development coefficient, and log population as the dominant covariate.

6 | Discussion

6.1 | Main Findings and Their Interpretation

AI-based crop-yield research aligns in part with agricultural scale and unevenly with selected capacity indicators; whether it aligns with research need cannot be resolved at the precision available. Harvested area is the one covariate that behaves the same way in every specification: the more land a country devotes to a crop, the more likely that country-crop system is to have been studied. The coefficient stays within a narrow band across the model variants, including crop and region fixed effects, and it is positive with an interval excluding zero in each of the five most studied crops estimated separately. Cropland extent accounts for a real share of where evidence exists, and a reading of the research map that ignored it would mistake agricultural structure for neglect.

Alignment stops short of proportionality. The rank correlation between fractional study counts and agricultural scale weakens once the crop dimension is restored, so agreement visible in national totals dissolves at the level where crops are grown and modelled, and research share rises with area at roughly a third of the rate proportionality would require (Section 5.2). Both the share comparison and the freely estimated elasticity confirm this pattern. Research effort per hectare is highest in mid-sized systems rather than the largest ones, which is invisible in raw counts, where the largest systems dominate. The departures run in both directions: the major cereals absorb more attention than the land they occupy, whereas oil palm is the clearest shortfall (Figure A1). Neither pattern is visible in national totals alone.

The association with research need is estimated much less precisely. The intensity association is positive under the scale-excluded index, but the result is sensitive to both index construction and the harvested-area restriction: it disappears under the eleven-component variant, and estimating the elasticity freely returns an interval containing zero. Imprecision of this kind does not show that research need is unrelated to research attention. The need association remains unidentified, and that is the limit of what these data support.

Capacity indicators are estimated more precisely than need but show no consistent directional pattern. R&D expenditure and internet access are positively associated with research intensity, whereas tertiary enrolment enters negatively and GDP per capita is imprecisely estimated. The mixed signs caution against interpreting the capacity covariates as a single latent dimension: what the estimates support is an association with particular capacity-related indicators, principally research financing, not with capacity in general. The expenditure association also moves across income-group splits, so it carries less weight than the area result.

The mismatch between a country's research standing and its need rank is spatially clustered more strongly than any other quantity examined, and it clusters in both directions where research output alone does not. Output on its own produces mostly Low–Low clusters, a pattern dominated by neighbouring countries with low research output; mismatch produces High–High clusters across western and northern Europe against Low–Low clusters concentrated in sub-Saharan Africa (Figure 5). Over- and under-representation relative to need are regional conditions shared with neighbours rather than isolated national cases. The practical implication runs against how research gaps are usually described. If mismatch were a national characteristic, it could be addressed country by country. Low-mismatch countries tend to be geographically adjacent, so research shortfalls are spatially concentrated rather than isolated.

6.2 | Geographic Representation and Transferability

Concentration matters if models do not transfer across settings. The transferability studies reviewed above answer in a consistent direction: apparent accuracy degrades as the evaluation domain moves away from the training domain, and the degradation is invisible under the validation designs most commonly reported [4, 5, 6].

TABLE 6 | Estimated quantities across the robustness suite

Estimated quantity	Baseline	Range across variants	Variants	Sign retained / interval excludes zero
Gini of country research counts, all countries	0.848	[0.771, 0.904]	13	13 of 13 / not applicable
Logit coefficient on log harvested area	0.723	[0.668, 0.838]	11	11 of 11 / 11 of 11
Global Moran's I on the mismatch measure	0.364	[0.190, 0.429]	13	13 of 13 / 13 of 13
Share of crop attention held by wheat, maize and rice	69.5%	67.9% under the machine-learning subset	2	not applicable
Logit coefficient on R&D expenditure	0.323	[−0.126, 2.375]	11	10 of 11 / 6 of 11
Logit coefficient on the research-need index	0.880	[−9.405, 1.081]	11	10 of 11 / 0 of 11
PPML coefficient on the research-need index	1.282	[0.114, 1.282]	4	4 of 4 / 2 of 4
Harvested-area elasticity, freely estimated	0.849	[0.755, 0.944]	1	unit elasticity rejected, $p = 0.002$

Baseline is the value in the main specification: the nine-component scale-excluded need index on the corrected coverage floor, observed-sample spatial statistics, and harvested area entered as an exposure offset in the intensity model. Range is the span across the variants listed in Appendix D, which include counting rules, document type, influential-country exclusions, period splits, the machine-learning subset, crop and region fixed effects, income-group splits, both need indices, alternative weight matrices, great-circle neighbours, alternative tie conventions, and the free-elasticity specification. The wide range on the need coefficient is driven by the low-income subsample, which contains 14 countries. The final column reports how many variants preserve the sign of the baseline estimate and how many return an interval excluding zero; no categorical stability label is assigned, because the thresholds that would define one were not fixed before the results were examined.

TABLE 7 | Determinants of research participation and research intensity

	Participation (logit)		Intensity, area offset		Intensity, free elasticity	
Covariate	Estimate	95% CI	Estimate	95% CI	Estimate	95% CI
Log harvested area	0.723 (0.051)	[0.622, 0.824]	offset		0.849 (0.048)	[0.755, 0.944]
R&D expenditure (% GDP)	0.323 (0.124)	[0.081, 0.565]	0.521 (0.122)	[0.281, 0.760]	0.451 (0.110)	[0.236, 0.667]
Research-need index	0.880 (0.957)	[−0.997, 2.756]	1.282 (0.583)	[0.139, 2.424]	0.982 (0.576)	[−0.148, 2.111]
Log GDP per capita	0.338 (0.350)	[−0.347, 1.023]	−0.045 (0.236)	[−0.507, 0.417]	0.031 (0.228)	[−0.416, 0.477]
Tertiary enrolment	0.213 (0.472)	[−0.712, 1.138]	−0.961 (0.427)	[−1.798, −0.124]	−0.842 (0.444)	[−1.713, 0.029]
Internet use	1.342 (1.018)	[−0.654, 3.337]	2.167 (0.831)	[0.537, 3.797]	1.831 (0.812)	[0.240, 3.423]
Log population	0.109 (0.108)	[−0.103, 0.321]	−0.107 (0.055)	[−0.215, 0.002]	0.016 (0.065)	[−0.111, 0.143]
Country-crop systems	1,799		1,792		1,792	
Countries	142		142		142	
Pseudo-R^2	0.365		—		—	

The participation column reports a logit for whether a country-crop system carries any eligible study. The two intensity columns report Poisson pseudo-maximum-likelihood estimates of the fractional study count: the first enters log harvested area as an exposure offset, fixing its elasticity at one; the second estimates that elasticity, which is 0.849 (95% CI 0.755 to 0.944) and rejects the unit restriction ($\chi^2_1 = 9.73$, $p = 0.002$). The research-need index is the nine-component scale-excluded measure; yield volatility enters through the index and is not a separate regressor. Standard errors, clustered by country, are in parentheses.

TABLE 8 | Average marginal effects on the probability that a country-crop system is studied

Covariate	AME	(SE)	95% CI	25th pct.	75th pct.	IQR effect (pp)
Log harvested area	0.0812	(0.0044)	[0.0725, 0.0898]	8.27	12.34	35.1
R&D expenditure (% GDP)	0.0362	(0.0138)	[0.0091, 0.0634]	0.18	0.94	2.8
Research-need index	0.0987	(0.1081)	[−0.1131, 0.3106]	0.25	0.76	4.8
Log GDP per capita	0.0379	(0.0396)	[−0.0398, 0.1156]	8.81	10.53	6.7
Tertiary enrolment	0.0239	(0.0529)	[−0.0797, 0.1275]	0.15	0.71	1.3
Internet use	0.1505	(0.1115)	[−0.0679, 0.3690]	0.55	0.91	5.6
Log population	0.0122	(0.0121)	[−0.0114, 0.0359]	15.76	17.82	2.6

Average marginal effects from the participation logit of Table 7, computed by the delta method with standard errors clustered by country. The final column reports the change in mean predicted probability, in percentage points, when the covariate moves from its 25th to its 75th percentile with all other covariates held at their observed values. Percentage-point changes are comparable across covariates; raw coefficients are not.

Those comparisons are made within fields and regions, and the present study does not estimate model transfer across countries. Spatially concentrated evidence does not by itself establish that models perform badly in settings the literature has not covered. The concern is narrower and measurable: of the 2,343 country-crop systems with a positive harvested-area denominator, 1,866 carry no eligible

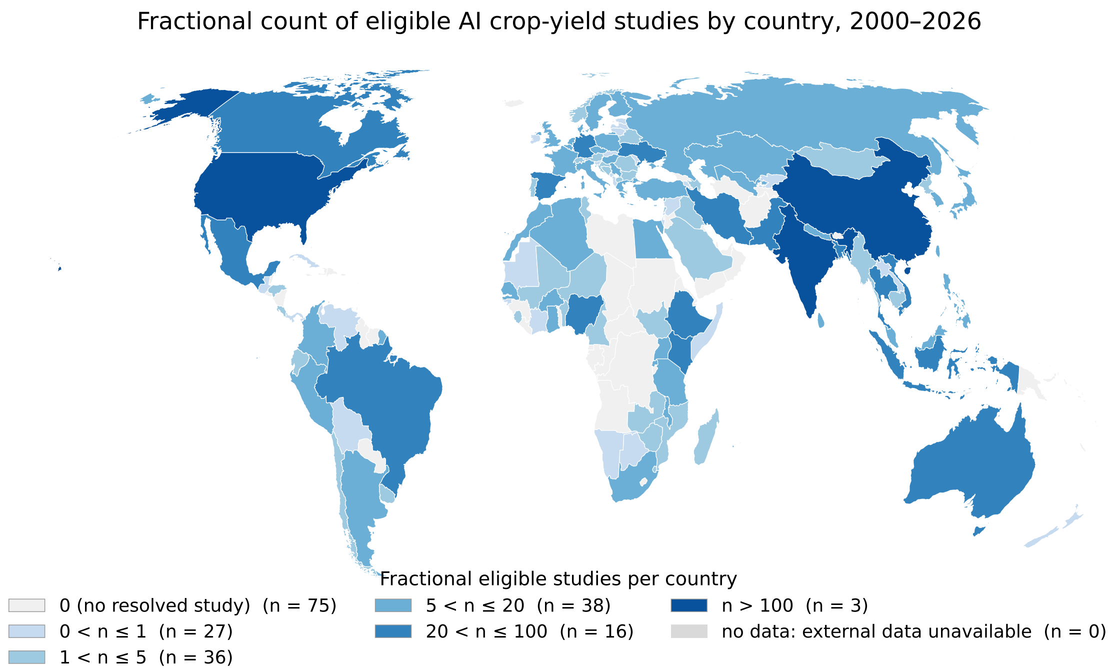

FIGURE 1 | Fractional AI/ML crop-yield research output by country. The map shows the 195 countries in the boundary layer; four of the 199 panel countries have no matching geometry and none of the four carries an eligible study, so 73 zero-study countries appear here against 77 in the full panel. Countries with no resolved eligible study are shown in the zero class and are distinguished from countries with no available data.

study, and those systems account for 19.1% of the panel's harvested area. Almost four-fifths of the world's cropping systems, covering close to a fifth of its cropland, have no in-domain evidence to degrade from. The concentration limits the geographic breadth of the evidence base; it does not demonstrate failure of any model, and this design could not detect such failure.

6.3 | Research Need versus Scientific Capacity

Research need and scientific capacity answer different questions, and the analysis keeps them apart: need describes where additional evidence would be useful, capacity where research can be produced. The observed geography is more consistent with differences in some dimensions of scientific capacity, particularly research financing, than with differences in need. Support for that reading is asymmetric: the R&D association is estimated with a standard error of about a tenth of a log-odds but moves across income-group splits, while the need association is imprecise for participation and, for intensity, positive only under a restriction the data reject. The available precision does not support a stronger conclusion.

The asymmetry matches what the agricultural research literature reports about the systems producing the work. In low- and middle-income countries the smallest agricultural research systems sit inside weaker surrounding institutions and thinner innovation environments [10], a constraint on producing research that operates independently of how much a country would gain from the evidence. Nin-Pratt [9] adds a caution: the conventional intensity ratio understates the research effort developing countries make, so low measured intensity should not be read as low effort, even though an investment gap close to half of R&D spending remains. Low output in the Low–Low cluster is therefore better read as a constraint on research production than as a revealed judgement about the value of the research, particularly because Rosegrant, Wong, Sulser, Dubosse, and Lybbert [11] estimate a central benefit–cost ratio of 33 for expanded agricultural R&D in the Global South.

Food-insecure countries receive about a fifth of global agricultural investment [19], a distribution that resembles the research pattern reported here. The resemblance is untested, because the analysis does not link investment flows to study locations. The Low–Low cluster should not be read as uniform either: returns to agricultural innovation investment vary by subregion within sub-Saharan Africa [20].

Agricultural scale on its own does not account for research geography either. Harvested area is associated with whether a system is studied but not with which large systems attract attention, because the departures from proportionality noted above occur among the largest producers.

6.4 | Implications for AI-Based Crop-Yield Research

Study location deserves to be reported as a matter of course. Author affiliation does not identify where a model was trained or evaluated, and fewer than half of the eligible studies here carry a usable empirical location (Appendix B). An explicit statement of study area would make the geography of this literature measurable without inference. Geographically independent validation is what would establish whether models generalise across geographic domains. The transferability studies above measure that gap within regions; extending the practice to settings where evidence is thin would turn a question about generalisability into a measured quantity.

Low research intensity relative to production or need marks where evidence is absent, not where funding should go. The mismatch

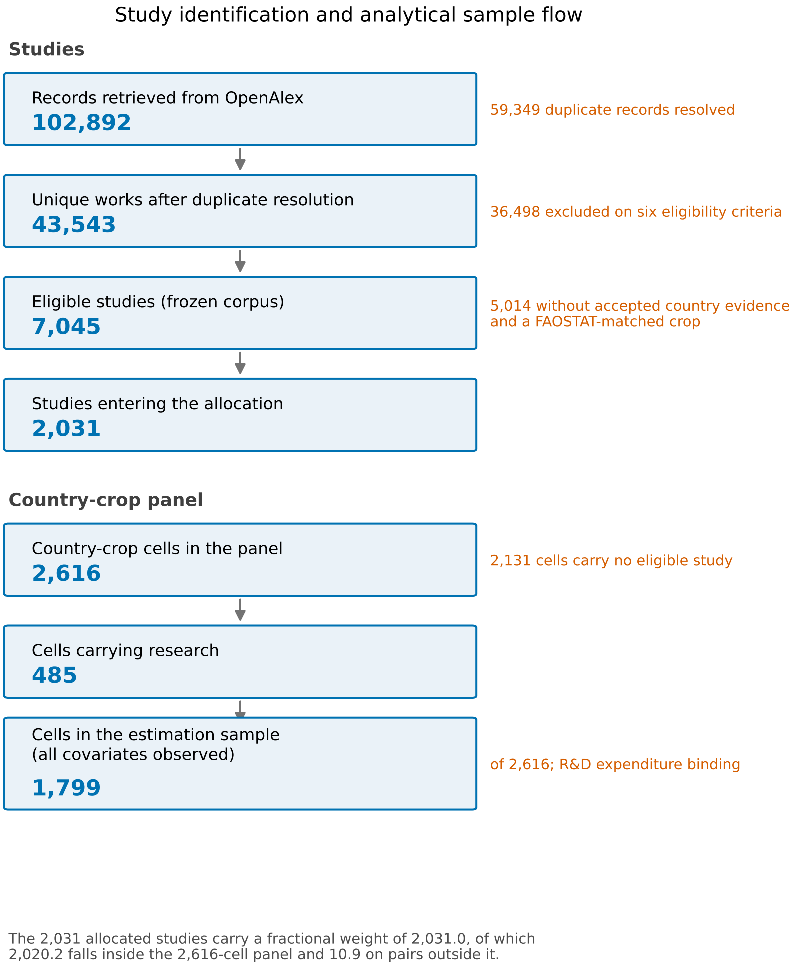

FIGURE 2 | Study identification and analytical sample flow. Counts on the right are the records surviving each stage; notes on the left of each arrow give what the stage removes. The fractional totals differ because 10.9 study-equivalents fall on country-crop pairs outside the 2,616-cell panel.

measure ranks evidence available against evidence that would be useful, and says nothing about the cost of producing it, the capacity to absorb it, or competing claims on research budgets. Reading the Low–Low cluster as an allocation rule would extend the measure past what it estimates.

Comparing counts against agricultural denominators is more informative than publication totals. Raw output confounds the size of a country's research system with the size of its agriculture. Agricultural denominators distinguish research concentration associated with agricultural scale from disproportionate allocation.

6.5 | Limitations

Location resolution bounds every geographic statement made here. Fewer than half of the eligible studies (43.4%) carry a usable empirical location, and about a quarter resolve to specific countries (Appendix B). The geographic patterns therefore describe the resolvable evidence base rather than every eligible publication. Coding accuracy was measured rather than assumed: allocation to countries is correct for 95.6% of an adjudicated probability sample (95% CI 90.7 to 98.0), but 62.0% of sampled unresolved records name a study area the coder did not recover, which projects to roughly a thousand studies. Resolved and unresolved studies differ on every observable tested (Table A6): resolved studies carry more citations (23 against 15), are more likely to have an abstract (93.9% against 80.2%), and are less often conference items (15.7% against 23.4%), while publication years differ by half a year. Resolved studies are also somewhat less likely to name an explicit machine-learning method (49.8% against 62.4%), so the resolvable base is not a methodologically narrower slice of the corpus. Against that, the countries named in the unrecovered records track the existing allocation closely by income group (high income 34.5% against 33.6%, low income 3.4% against 3.6%) and by mean need percentile (0.380 against 0.355), so on the dimensions that can be observed the under-resolution does not look geographically selective. It may still be selective on dimensions that cannot. The two error rates point in opposite directions and matter differently. High allocation accuracy means the countries shown on the map belong there, so the concentration is not manufactured by misassignment. Low recall means the map is a thinned version of the true distribution, which leaves counts too low everywhere but leaves their relative pattern intact if the thinning is close to random, as the income and need comparisons suggest. The concentration result is more secure than the absolute counts that underlie it.

Screening was automated, with a false-inclusion rate of 10.0% and a false-exclusion rate of 2.2% in the largest exclusion stratum, both estimated from 85 adjudicated records whose interval upper

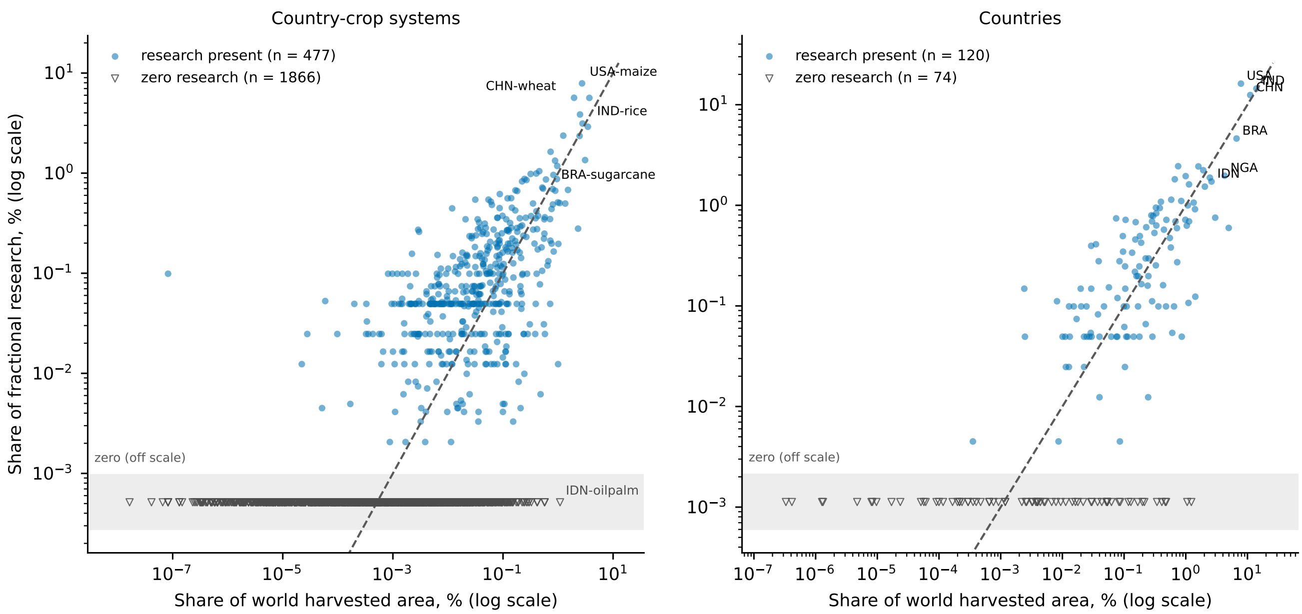

FIGURE 3 | Share of fractional research against share of world harvested area, for country-crop systems (left) and countries (right). Both axes are logarithmic; the dashed line marks equal shares; units with no resolved study carry no position on a logarithmic axis and are drawn as hollow triangles on the shaded band below the plotting range, which is off scale rather than a small positive value.

bounds reach 0.236 for false inclusion and 0.116 for false exclusion (Appendix A). Counts carry classification error that is estimated but not tightly bounded, and a cell recorded as zero could reflect a missed or unresolved study rather than an absent literature. Both the screening and the coding validations were adjudicated by model-assisted review against the same criteria rather than by independent human reading, so they measure the extractor against its own evidence base and require human confirmation before they can be treated as settled.

Research need is measured at country level while the analytical unit is the country-crop system, so the need index cannot vary across crops within a country. The need coefficient is identified from between-country variation alone, which is part of why it is imprecise, and a need measure varying by crop might behave differently. The covariates are the most recent observations rather than a common year. The models compare the accumulated 2000–2026 evidence stock with current conditions, and are read as measures of present alignment rather than as historical determinants of research production. The median reference year is 2023 or later for every covariate, although R&D expenditure is older in low- and lower-middle-income countries, with medians of 2018 and 2019 against 2023 for the two higher-income groups (Table A7). Missing capacity data compounds the restriction: R&D expenditure is the sparsest covariate (Table 3), and the models are estimated where capacity indicators are observed. Observation is itself capacity-related, so the countries least represented in the estimation sample are disproportionately those the need question concerns most.

Study counts measure research attention rather than research quality or societal value, and publication output is at best a crude indication of underlying research capacity [8]. No specification identifies why a system is studied, and reverse causation is not excluded. Spatial results depend on the neighbour structure imposed on a country set containing many islands, though the mismatch clustering holds across the neighbour definitions, the common connected subsample, and both need indices.

7 | Conclusion

Whether AI-based crop-yield research is distributed in proportion to where crops matter, where evidence is most needed, or where research can most readily be produced is an empirical question that publication counts cannot answer on their own. The analysis codes each study's empirical location from its reported study area rather than author affiliations, allocates it fractionally across country-crop systems, and compares the result against harvested area, production, research need, and capacity indicators.

Geographic coverage is limited: 122 of 199 countries carry any eligible study, 485 of 2,616 country-crop systems contain research, and wheat, maize, and rice absorb 69.3% of crop attention. Harvested area shows the largest and most stable association with research participation among the covariates examined, and it is the only one whose sign and precision hold across every specification tried. Correspondence with agricultural scale is partial, weakening once the crop dimension is restored. National research and development spending is associated with both participation and intensity, though the association moves across income groups, and the other capacity-related indicators show no consistent directional pattern. The need association is too imprecise to resolve, which limits what can be concluded about need rather than settling it. The mismatch between research standing and need is itself geographically patterned, and more sharply than the counts it is built from: neighbouring countries share it, positively across western and northern Europe and negatively across much of sub-Saharan Africa.

Counting publications by the country of their authors, or counting them at all without a denominator, cannot separate concentration

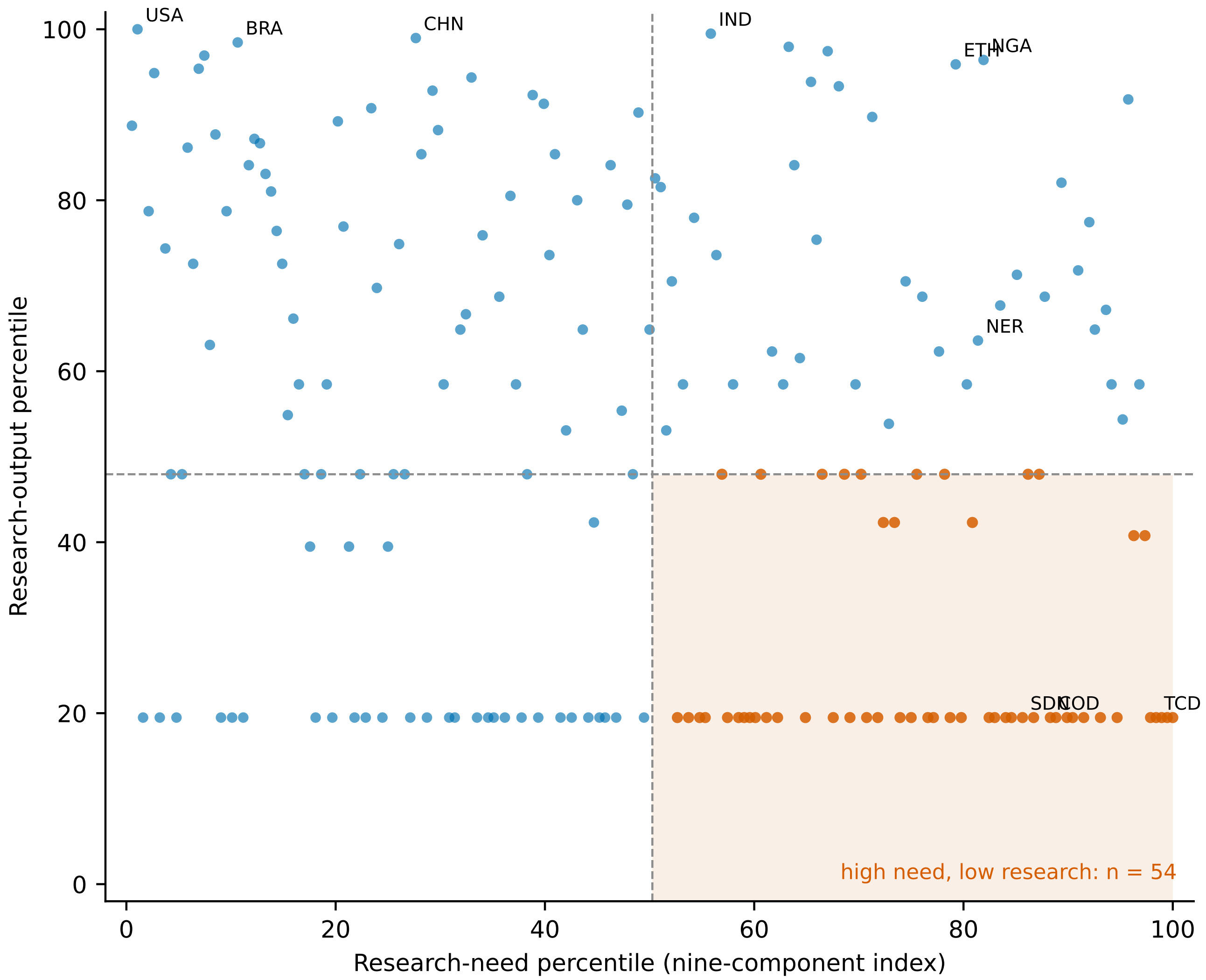

FIGURE 4 | Country research percentile against research-need percentile. 188 countries with a computable nine-component need index. The shaded quadrant holds the 54 countries at or above the median for need and at or below the median for research.

that follows agriculture from concentration that departs from it. Placing study-location research attention alongside agricultural scale, research need, and scientific capacity in one country-crop frame separates them, and a large share of the world's production systems carries no evidence at all. Nothing in the analysis establishes that models fitted in well-studied settings perform poorly elsewhere, because cross-country predictive transfer is not tested here. The uneven geographic coverage, and the large number of country-crop systems with no resolved evidence, limit the extent to which this literature can be treated as representative of global cropping systems.

Widening that coverage depends on practices the field can adopt directly. Study locations stated explicitly would remove the inference now needed to place more than half the corpus. Validation designs that hold out geography rather than observations would establish whether models generalise across geographic domains beyond the settings that produced them. Neither practice substitutes for the capacity constraints that shape where agricultural research can be produced, and those constraints fall hardest on the country-crop systems where the shortfall relative to need is largest.

Author Contributions

Prasamsha Poudel and Saugat Khanal contributed equally to this study.

Conflicts of Interest

The authors declare no conflicts of interest.

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Data Availability Statement

The bibliographic corpus was assembled from the public OpenAlex database. Agricultural production and harvested-area data are from FAOSTAT, research-need components from FAOSTAT food security, macroeconomic, employment, and temperature domains, scientific-capacity indicators from World Bank Open Data, and country boundaries from Natural Earth; all sources are publicly available. The

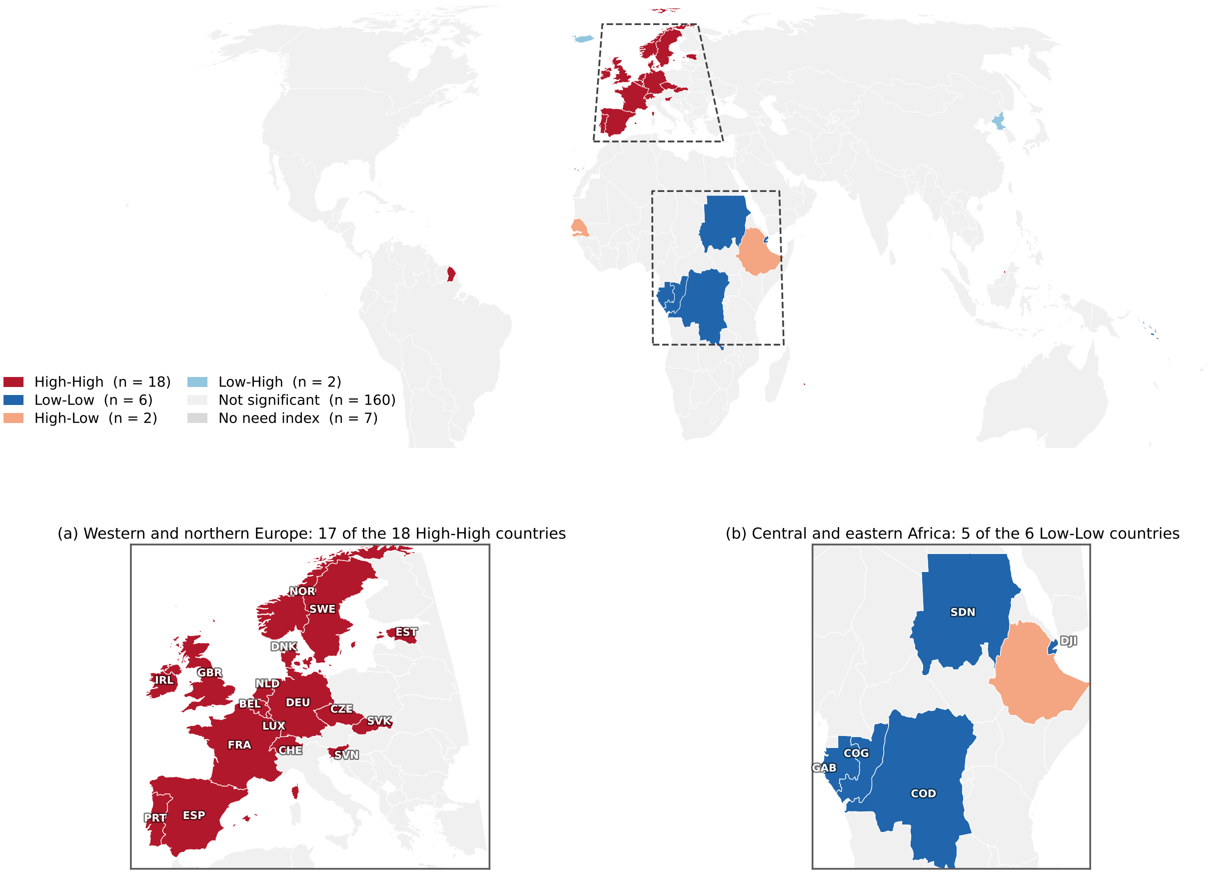

FIGURE 5 | Local Moran clusters of the research-need mismatch measure. Categories are assigned on Benjamini–Hochberg adjusted pseudo p-values from 9,999 permutations, computed over the 188 countries with an observed mismatch value under the primary six-nearest-neighbour weights. The dashed rectangles mark the extents enlarged in the two panels below, which label every clustered country. Panel (a) shows 17 of the 18 High–High countries, the exception being Brunei Darussalam; panel (b) shows five of the six Low–Low countries, the exception being Solomon Islands.

coded study-level dataset, the derived country-crop panel, the analysis code, and the outputs behind every table and figure are available at https://github.com/Saugat2000/ai-crop-yield-research-alignment.

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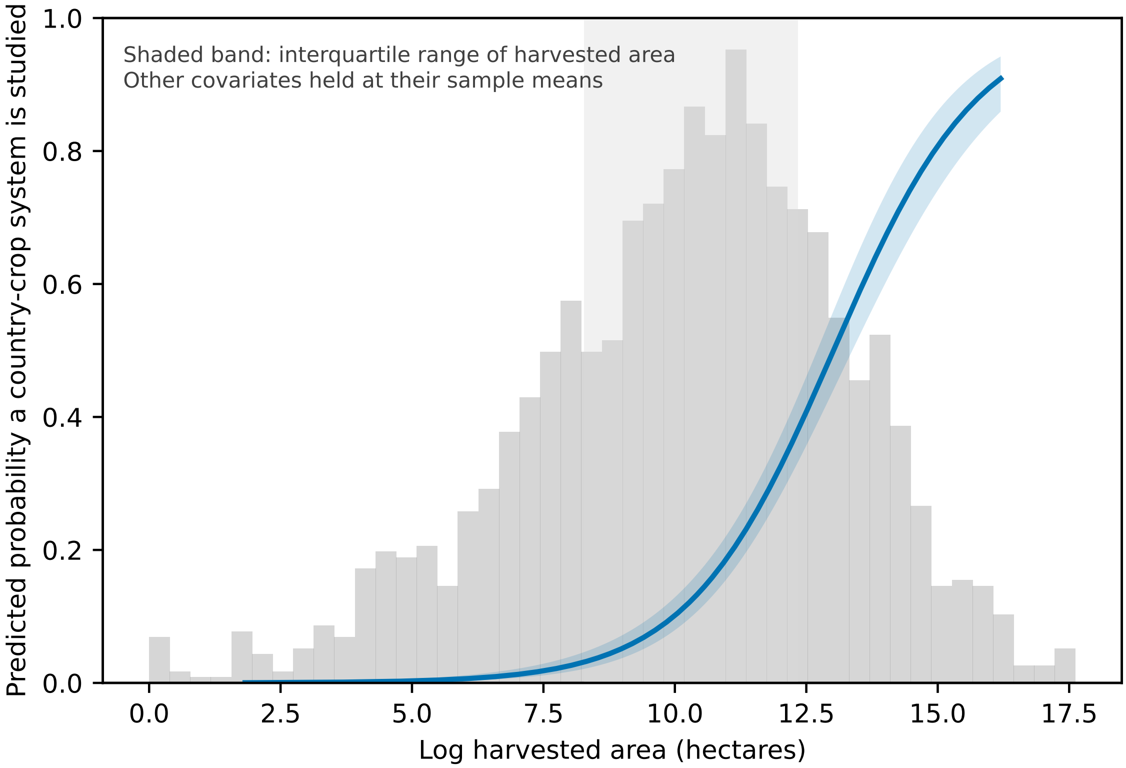

FIGURE 6 | Predicted probability that a country-crop system carries research, as a function of log harvested area, with all other covariates held at their sample means. The band is a 95% confidence interval; the histogram shows the distribution of log harvested area across the estimation sample.

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APPENDIX

Appendix A | Search, Screening, Deduplication, and Corpus Validation

Retrieval ran as two passes over OpenAlex, one restricted to journal and preprint document types and one to conference papers and abstracts, executed as 86 query expressions. All 86 completed. Retrieval returned 102,892 records. Removing duplicate identifiers left 45,845 unique records. Records sharing a DOI, a normalised title, or an author-year signature were then resolved into single works, yielding 43,543 unique works. Ambiguous groupings were split rather than merged where the evidence for merging was weak.

Screening applied six eligibility criteria to the title and abstract of all 43,543 works: the study concerns an agricultural crop; crop yield is the modelled, predicted, or estimated outcome; the method is genuinely data-driven or machine-learning based; the work reports an empirical application; the document type is eligible; and the publication date falls from 2000 onward. Process-based crop simulation without a learned component fails the third criterion.

Screening used an automated rule set rather than human reading, and the corpus records the decision method. Exclusions distinguish those resting on affirmative evidence from those resting on the absence of evidence, because the second class can lose an eligible study irrecoverably. Error rates were estimated from stratified probability samples adjudicated against the same six criteria. Eighty-five records were adjudicated across three strata: 45 drawn from the largest exclusion stratum of 8,677 records, and 20 each from two inclusion strata of 4,143 and 2,902 records, with design weights of 99.7, 82.9, and 58.0. The false-inclusion rate is 10.0% and the false-exclusion rate 2.2%. The point estimates satisfy the pre-specified ceilings; the 95% interval upper bounds do not, reaching 0.236 and 0.360 for false inclusion against a ceiling of 0.150 and 0.116 for false exclusion against a ceiling of 0.050, because the adjudicated samples are too small to bound a rate of that size. Three adjudications returned an uncertain verdict. Only one exclusion reason was sampled, so the false-exclusion estimate does not generalise to the other exclusion strata. The implied 257 ineligible studies inside the corpus are reported rather than removed, and the screening thresholds were not altered after the fact. Adjudication was model-assisted review against the recorded evidence, not independent human reading, and is reported on that basis (Table A2).

The corpus contains 7,045 eligible studies: 4,842 journal articles, 1,377 conference papers, 252 preprints, 249 book chapters, 185 dissertations, 117 conference abstracts, and 23 reports, giving 1,494 conference papers and abstracts combined. Abstracts are present for 5,927 studies. A deterministic keyword screen of titles and abstracts finds explicit machine-learning method terms in 62.6% of the corpus, classical-regression terms alone in 9.6%, process-model terms alone in 2.9%, and no method keyword in 24.9%, mostly where no abstract is available. Resolving the corpus into model families gives deep learning in 29.6% of records, tree ensembles in 24.4%, support-vector and kernel methods in 10.4%, other machine-learning methods in 40.8%, classical statistical methods in 17.4%, and process-based models in 6.4%; the families overlap because studies compare several. Rerunning the main analyses on the 4,261-study subset carrying an explicit machine-learning term changes none of them materially: the all-country Gini moves from 0.843 to 0.868, the crop Gini from 0.719 to 0.714, the top-three crop share from 69.3% to 67.7%, the participation coefficient on log harvested area from 0.713 to 0.707, the intensity coefficient on research and development expenditure from 0.522 to 0.508, and Moran's I on the mismatch measure from 0.353 to 0.291, with $p = 0.0001$ throughout. The breadth of the eligibility definition does not drive the reported geography. The corpus was frozen on 13 August 2026 and is fixed by the SHA-256 checksum `98c132b0...26024c`; every dataset used in the analysis derives from that frozen set.

Appendix B | Study-Location and Crop Coding Protocol

Study location is coded from the study's own account of where its agricultural data originated. A country name alone is never sufficient evidence. The name must appear in a context that positively asserts an empirical study area, such as a statement that field trials were conducted or yield records obtained in a named place. Country names occurring within an institution name, a funder, a cultivar name, or a citation are evidence about authorship or provenance rather than study area, and are rejected with the reason recorded.

Each resolved record carries the cue type that produced the assignment. The value `study_area` marks an explicit study-area statement. The value `locative_only` marks an assignment resting on a weaker locative phrase; such assignments are capped at medium confidence and flagged so that they can be dropped in sensitivity analysis. A citation-context filter prevents a country named only as the subject of previously published work from being read as a study area.

Three outcomes are held distinct from a resolved single country. Studies whose scope is global or continental receive a scope label and no single country code, because assigning a global study to one country would create an observation the study does not support. Ambiguous toponyms, including Georgia, Jordan, and Chad, are a country in one reading and a subnational unit, a person, or a common noun in another; these require a disambiguating cue and otherwise return an ambiguous result. Studies naming no study area return unresolved, recorded as a distinct state rather than as an absence of research.

Location coding returns five mutually exclusive states: 1,821 studies resolve to a single country, 56 to several named countries, 915 are global in scope, 268 regional, and 3,985 unresolved, summing to the 7,045-study corpus; 43.4% carry some usable location signal. Of the country-resolved studies, 869 rest on the weaker locative cue. For crops, 4,454 studies name at least one identified crop, 654 more than one, and 2,364 none.

Location and crop coding were produced by an automated, model-assisted extractor. Every record stores the value, the verbatim evidence text, the cue type, a confidence level, and a verification status; all 7,045 records are stored as unverified, and none was confirmed by independent human reading.

Coding accuracy was estimated from a stratified probability sample of 210 records drawn under seed 20260831 across seven strata spanning study-area and locative cues, multi-country, global-scope, regional, and unresolved records, with design weights from 2.8 to 184.6. Verdicts were assigned by model-assisted adjudication against the title and abstract, which is a first pass requiring human confirmation rather than independent review. Two quantities differ and are reported separately. Allocation accuracy, meaning that the countries receiving research weight are the countries the study examines, is 90.4% (123 of 136 determinable records, 95% CI 84.3 to 94.3); 0.7% of records are allocated wholly to the wrong country, and 8.8% miss or add a country, almost all in the multi-country stratum. Scope labelling is weaker: the global label is often triggered by generic wording such as global food security or world population, and 11 of 20 sampled regional records name no study area and should be unresolved. Scope-label errors mostly leave allocation intact, because allocation runs on accepted country mentions rather than the scope label. The validation identified three reproducible false-positive patterns, each arising from a country name matched inside a larger toponym or an agro-ecological zone name: all six records mentioning New South Wales were coded to the United Kingdom, all three mentioning the Guinea savanna were coded to Guinea, and all four records coded to Georgia refer to the United States state. Every affected record was read individually and repaired, and the thirteen corrections are listed with their evidence in the replication materials. After repair, Guinea and Georgia carry no eligible study, the United Kingdom holds 21 rather than 27, and Australia, Ghana, Nigeria, and the United States gain the weight removed from them. The allocation reported throughout this paper is the repaired one, and the accuracy in Table A2 is measured on it. Recall is the weaker property: 62.0% of sampled unresolved records (95% CI 48.2 to 74.1) name a study area the coder did not recover, projecting to roughly 1,000 of the 3,985 unresolved studies. Those unrecovered records track the existing allocation closely by income group and by mean need percentile, so on observable dimensions the under-resolution does not appear geographically selective. Crop coding precision is 89.4% (152 of 170, 95% CI 83.9 to 93.2). Table A2 reports the design and the error taxonomy.

Allocation uses each study's accepted country evidence: the resolved country for single-country studies, and otherwise every country the coder accepted as an explicit empirical mention. That rule admits some studies whose overall scope label is global or unresolved but which name concrete study areas in their titles or abstracts. Under it, 2,031 studies carry both accepted country evidence and a FAOSTAT-matched crop: 1,347 single-country, 47 multi-country, 343 global-scope, and 294 unresolved-scope. Each study contributes a total weight of one, divided equally across the country-crop combinations it covers, giving 2,966 study-country-crop records whose weights sum exactly to 2,031. Of that weight, 10.9 study-equivalents fall on country-crop pairs with no FAOSTAT denominator row and sit outside the 2,616-cell panel, which carries 2,020.2. The allocation rule does not drive the results: excluding the global-scope allocations leaves the all-country concentration Gini at 0.847, and restricting allocation to studies whose scope names specific countries gives 0.839. Full counts, in which a study contributes one to every cell it covers, are stored alongside the fractional counts and used only for descriptive comparison.

Appendix C | Variable Definitions, Sources, Coverage, and Index Construction

Agricultural scale

The FAOSTAT crop extract covers 1990 to 2024 and contains 111,378 country-crop-year observations, recording harvested area in hectares, production in tonnes, and yield in tonnes per hectare. Where FAOSTAT does not report yield directly, yield is derived as production divided by harvested area, and the derivation is recorded in the source field.

The panel universe is defined as follows. Take the 199 countries and 25 crops carried through the identifier crosswalk, giving 4,975 possible combinations. Retain the combination where FAOSTAT reports the crop in that country for any year from 1990 to 2024, which gives 2,590 cells, and add 30 cells for Guadeloupe, French Guiana, Martinique, and Réunion, which the analytical layer holds separately but FAOSTAT reports inside France. Four cells present in FAOSTAT (Jordan-pulses, Chad-yam, Saint Vincent and the Grenadines-sugarcane, Yemen-sugarcane) fall outside the panel through the crosswalk. The result is 2,616 cells. Of these, 109 carry no FAOSTAT area, production, or yield value for the reference period and support no denominator-based comparison; none of the 109 carries research, so restricting the panel to the 2,507 cells with an observation moves the share of studied systems from 18.5% to 19.3%.

Each panel cell takes the mean of the five reference years 2018 to 2022. The number of reference years actually observed is stored with each cell: 2,464 of the 2,616 cells draw on all five. Production values are present for 2,505 cells, of which 2,345 are strictly positive; harvested area is present for 2,504 cells, of which 2,343 are positive. Cells carrying a recorded zero are retained and stay distinct from cells with no observation.

Country identifiers are harmonised through a crosswalk linking FAOSTAT M49 codes, ISO alpha-3 codes, World Bank names, and the alpha-2 codes OpenAlex uses for affiliations. An unmatched identifier is logged rather than dropped. FAOSTAT aggregate areas, including World and continental groupings, are flagged and excluded from country-level analysis.

Crops are standardised to 25 items spanning cereals, roots and tubers, pulses, oilseeds, sugar crops, and the fibre and beverage crops appearing in the corpus. Several items group closely related FAOSTAT commodities that the study literature does not distinguish, including banana and plantain, oat and rye, and coffee and cocoa. Crops named in eligible studies but lacking a separate FAOSTAT item, teff among them, cannot receive a production denominator and are recorded as such rather than folded into a residual category.

Research-need components and index construction

The research-need index combines the nine non-scale indicators listed in Table 3; the two agricultural-scale shares are recorded with the need data but excluded from the index because they are the paper's importance measures. Indicators are recorded with their declared direction, converted to percentile ranks, and combined into rank-aggregation, equal-weighting, entropy-weighting, and first-principal- component indices, plus the eleven-component variant used for sensitivity. Each index is also stored as a percentile on a zero-to-one scale. Component coverage varies and is stored for every country. The floor is five of the nine primary components, and it operates in three cases: below the floor no index is computed; at or above it but incomplete, the index is the mean of the components observed; complete, the index uses all nine. That rule indexes 188 of the 199 countries.

The floor counts the nine need components only. An earlier version counted all eleven recorded indicators, which include the two agricultural-scale shares. Because those two shares are observed for every country, each country received two components toward the threshold without carrying any need information, and five countries (the Cook Islands, Micronesia, Saint Kitts and Nevis, Singapore, and Tuvalu) were indexed on between one and four observed need components. The index value excluded agricultural scale while the inclusion rule did not, so the floor was moved onto the nine primary components. The change is immaterial to every reported quantity. Holding the repaired allocation fixed and varying only the floor, the mismatch clustering statistic moves from 0.366 under the eleven-indicator rule to 0.364 under the nine-component rule, and the high-need low-research quadrant from 56 countries to 54. The eleven-indicator variant is retained for comparison. The nine components are weakly related rather than interchangeable: Cronbach's alpha over the harmonised ranks is 0.669, the mean pairwise Spearman correlation is 0.243 (Figure A4), and the first principal component explains 41.9% of variance, so the index summarises several distinct dimensions rather than one latent construct.

Scientific-capacity indicators

Scientific capacity uses six World Bank indicators retrieved through the World Bank API and verified against the API's own record counts. Each country takes the most recent non-missing observation, and the year of that observation is stored beside the value so that staleness remains visible. Population required repair: the initial retrieval was truncated by an API paging limit, losing 1,540 observations across 42 entities, and the series was re-fetched and verified before use. All other indicators reconciled exactly against the API counts. Missing capacity values are not imputed, so models including research and development expenditure estimate on a smaller sample than models excluding it.

Spatial layer

Country geometries are the Natural Earth admin-0 boundaries in geographic coordinates on the WGS 84 datum (EPSG:4326). The analytical layer holds 195 countries, 93 represented as single polygons and 102 as multipolygons. Each record carries the FAOSTAT M49 code, ISO alpha-3 code, World Bank name, region, and income group, together with a polygon centroid and a land area computed on the Equal Earth projection. Nearest neighbours are found among those polygon centroids after projection to Equal Earth, using planar Euclidean distance, and every matrix is row standardised. Equal Earth preserves area rather than distance, so the statistics were recomputed with neighbours found on true great-circle distance: the neighbour set changes for 86 of 195 countries, and Moran's I on the mismatch measure is 0.364 against 0.359 at $k = 6$, 0.373 against 0.364 at $k = 4$, and 0.350 against 0.333 at $k = 8$, with local clusters of 16 High–High and 5 Low–Low against 17 and 6. Aggregate and historical entities are excluded and the reason for each exclusion is recorded. Four panel territories have no matching boundary; all four carry no eligible study.

Appendix D | Robustness and Sensitivity Specifications

Table A1 lists the specifications behind Section 4.6. Concentration variants re-compute the distribution of research attention across countries; model variants re-estimate the participation specification of Equation (5); spatial variants re-compute Equation (7).

Two estimation limitations are recorded here rather than in the body. The negative binomial model of intensity converged but did not return standard errors under the clustered covariance estimator, so it informs the overdispersion question only and no interval is read from it. A hurdle specification separating participation from positive counts was fitted during model development, but its coefficients were not retained in the reported output and it is not used.

Appendix E | Supplementary Figures

Figure A1 accompanies Section 5.1, Figure A2 accompanies Section 5.4, where the corresponding estimates are reported in Table 7, and Figures A3 and A4 accompany Sections 5.6 and 4.3.

TABLE A1 | Robustness and sensitivity specifications

Check	Main specification	Alternative	Purpose	Sample affected
Counting rule	Fractional, Eq. (1)	Full counting	Test whether concentration depends on how multi-country studies are divided	All studies
Attention weighting	Unweighted study count	Citation-weighted count	Test whether weighting by attention received changes the distribution	All studies
Document type	Journal and conference	Journal only; preprints excluded	Test whether conference and preprint literature drives the pattern	1,712 studies journal-only
Location confidence	All resolved locations	Locative-cue cells excluded; low-confidence excluded	Test whether weaker location coding drives the pattern	Cells resolved only by locative cue
Influential countries	All countries	Exclude USA; China; both; and with India	Test whether the largest contributors produce the concentration	Up to three countries
Period	Full 2000–2026	2000–2014; 2015–2019; 2020–2026	Test whether the pattern is specific to one period	All studies
Screening error	Frozen corpus	Adjustment for the measured false-inclusion rate	Bound the effect of screening error on concentration	Whole corpus
Need index	Rank aggregation	Equal weighting; entropy weighting; first principal component	Test whether the need result depends on aggregation	All indexed countries
Income group	Pooled	Low, lower-middle, upper-middle, high separately	Test whether associations differ by income	Estimation sample split
Spatial weights	kNN, $k = 6$	$k = 4$; $k = 8$; queen; distance band	Test whether spatial results depend on the neighbour definition	Queen drops 43 countries
Neighbour distance	Equal Earth planar	Great-circle distance	Test whether an equal-area projection distorts the neighbour sets	86 of 195 neighbour sets change
Rank ties	Average ranks	Minimum; maximum; dense; standardised log count	Test whether mismatch clustering depends on the tie convention	68 countries tied at zero
Need-index coverage floor	Five of eleven recorded indicators	Five, seven, or nine of the nine primary components	Test whether thinly measured countries drive the need result	188 countries at the corrected floor
Area elasticity	Fixed at one by the offset	Freely estimated, with a Wald test	Test the exposure restriction in the intensity model	1,792 cells
Method definition	Full eligible corpus	Studies naming an explicit machine-learning method	Test whether eligibility breadth drives the geography	4,261 studies

Sample affected refers to the units whose inclusion or weighting changes relative to the main specification.

TABLE A2 | Validation of screening, study-location coding, and crop coding

Stratum	Records	Sampled	Weight	Correct	Rate	95% CI
Panel A. Scope and country both correct, by stratum						
Global scope, countries accepted	343	30	11.4	0	0.000	[0.000, 0.114]
Multiple named countries	56	20	2.8	8	0.400	[0.219, 0.613]
Regional scope	268	20	13.4	5	0.250	[0.112, 0.469]
Single country, locative cue	859	45	19.1	34	0.756	[0.613, 0.858]
Single country, study-area cue	962	45	21.4	43	0.956	[0.852, 0.988]
Unresolved scope, countries accepted	294	30	9.8	2	0.067	[0.018, 0.213]
Unresolved, not allocated	3,691	20	184.6	11	0.550	[0.342, 0.742]
Panel B. Accuracy of the countries that receive research weight						
Countries receiving weight are correct				130	0.956	[0.907, 0.980]
Allocated wholly to the wrong country				1	0.007	[0.001, 0.040]
Allocation misses or adds a country				5	0.037	—
Panel C. Other quantities						
Location coding accuracy, single-country study-area cue				43	0.977	[0.882, 0.996]
Location coding accuracy, single-country locative cue				34	0.971	[0.855, 0.995]
False-unresolved rate (study area present but not coded)				31	0.620	[0.482, 0.741]
Crop coding precision (records with a determinable crop)				152	0.894	[0.839, 0.932]
Panel D. Screening validation						
Largest exclusion stratum	8,677	45	99.7	44	0.978	[0.884, 0.996]
Inclusion, all criteria affirmative	4,143	20	82.9	19	0.950	[0.764, 0.992]
Inclusion, empirical marker not named	2,902	20	58.0	14	0.700	[0.481, 0.855]

Panel A applies a strict rule: a record counts as correct only if the scope label and the country assignment both match the title and abstract. That rule explains the low rates for the global-scope and regional strata, where the scope label was triggered by generic wording such as global food security while the countries the study names were still allocated correctly. Panel B reports the quantity the analysis depends on, namely whether the countries that receive research weight are the countries the study examines; it is the statistic cited in Section 3.2 and it is higher than Panel A precisely because scope-label errors do not misdirect weight. Panel B is computed after the three systematic coding errors described in Appendix B were repaired. Intervals are Wilson intervals. All adjudication was model-assisted review against the recorded evidence rather than independent human reading, and requires human confirmation before it is treated as settled.

TABLE A3 | Specification checks on the participation and intensity models

Covariate	Baseline participation	Crop FE participation	No R&D, expanded	Intensity (offset)	Intensity, free area
Log harvested area	0.723	0.668	0.677	—	0.849
R&D expenditure (% GDP)	0.323	0.310	—	0.521	0.451
Research-need index	0.880	0.607	0.684	1.282	0.982
Log GDP per capita	0.338	0.320	0.635	−0.045	0.031
Tertiary enrolment	0.213	0.101	0.520	−0.961	−0.842
Internet use	1.342	1.064	0.362	2.167	1.831
Log population	0.109	0.309	0.187	−0.107	0.016
Cells	1,799	1,799	2,161	1,792	1,792

Coefficients only; intervals are in the replication outputs. Crop fixed effects identify the harvested-area association from variation across countries within a crop. The expanded column drops research and development expenditure, which restores 33 countries and 362 systems. The final column estimates the harvested-area elasticity rather than fixing it at one: the estimate is 0.849 (95% CI [0.755, 0.944]), and a Wald test rejects unit elasticity ($\chi^2_1 = 9.73$, $p = 0.0018$). Standard errors are clustered by country.

TABLE A4 | Nested block models of research participation

Model	Covariates	Log-likelihood	McFadden R^2	AIC	BIC
Constant only	0	-1007.5	0.000	2017.0	2022.5
Agricultural scale	1	-699.1	0.306	1402.1	1413.1
Scale + research need	2	-656.8	0.348	1319.5	1336.0
Scale + capacity indicators	6	-635.9	0.369	1285.8	1324.2
Scale + need + capacity	7	-634.6	0.370	1285.3	1329.2
Likelihood-ratio tests					
Comparison	χ^2	df	p	Δ McFadden R^2	
Need added to scale	84.59	1	<0.001	0.0420	
Capacity added to scale	126.33	5	<0.001	0.0627	
Capacity added to scale and need	44.24	5	<0.001	0.0220	
Need added to scale and capacity	2.50	1	0.114	0.0012	

Participation logit on the estimation sample, standard errors clustered by country. Capacity indicators are research and development expenditure, log GDP per capita, tertiary enrolment, internet use, and log population. Differences in fit describe how much each block adds to in-sample explanation and are not evidence of causal importance. The corresponding comparison for the intensity model, scaled by the estimated dispersion, distinguishes neither block.

TABLE A5 | Country-level models of research output

Covariate	Log study count		Studies per hectare (PPML)	
	Estimate	95% CI	Estimate	95% CI
Log harvested area	0.138	[0.07, 0.20]	−0.463	[−0.62, −0.31]
R&D expenditure (% GDP)	0.020	[−0.14, 0.18]	0.282	[0.07, 0.49]
Research-need index	−1.107	[−2.09, −0.13]	0.590	[−0.57, 1.75]
Log GDP per capita	−0.024	[−0.35, 0.30]	0.252	[−0.21, 0.72]
Tertiary enrolment	−0.202	[−0.80, 0.39]	−0.445	[−1.34, 0.45]
Internet use	1.083	[0.03, 2.14]	0.714	[−0.93, 2.36]
Log population	0.450	[0.33, 0.57]	0.354	[0.17, 0.54]
Countries	142		142	

Countries are the unit. The left column regresses the log of one plus the country's fractional study count on the same covariates as Table 7; the right column is Poisson pseudo-maximum likelihood with total harvested area as an exposure offset. Heteroskedasticity-robust standard errors. Estimates describe association, not effect.

TABLE A6 | Resolved versus unresolved studies

Measure	Resolved (2,031)		Unresolved (5,014)		Statistic	p
	Mean	Median	Mean	Median		
Publication year	2020.6	2022	2021.1	2023	4,567,372	<0.001
Citation count	23.0	4	15.4	2	5,811,177	<0.001
Percentages						
Journal article	75.9		65.8		68.2	<0.001
Conference paper or abstract	15.7		23.4		51.2	<0.001
Preprint	3.3		3.7		0.3	0.557
Other document type	5.0		7.1		9.8	0.002
Abstract available	93.9		80.2		204.8	<0.001
Explicit machine-learning method term	49.8		62.4		93.2	<0.001

Resolved studies are the 2,031 carrying accepted country evidence and a FAOSTAT-matched crop, which enter the allocation; the remaining 5,014 do not. Continuous measures are compared by Mann–Whitney U and percentages by chi-square on the two-by-two table. With samples of this size, small differences reach conventional significance: the publication-year difference is half a year, whereas the citation and abstract-availability differences are larger. The comparison characterises observable selection and does not establish that non-resolution is geographically random.

TABLE A7 | Covariate vintage, dispersion, and spatial diagnostics

Panel A. Reference year of the R&D expenditure observation

Income group	Countries	Median	Earliest	Latest
High income	54	2023	2015	2024
Low income	14	2018	2005	2024
Lower middle income	31	2019	2002	2024
Upper middle income	43	2023	1999	2024

Panel B. Intensity model: clustered against quasi-Poisson standard errors

Covariate	Estimate	SE (clustered)	SE (quasi-Poisson)	Ratio	p (clust.)	p (quasi)
R&D expenditure (% GDP)	0.521	0.122	0.598	4.9	<0.001	0.384
Research-need index	1.282	0.583	3.578	6.1	0.028	0.720
Log GDP per capita	−0.045	0.236	1.375	5.8	0.849	0.974
Tertiary enrolment	−0.961	0.427	2.872	6.7	0.024	0.738
Internet use	2.167	0.831	5.018	6.0	0.009	0.666
Log population	−0.107	0.055	0.290	5.3	0.054	0.714

Panel C. Lagrange Multiplier tests on country-mean residuals, $k = 6$

Test	Participation logit		Intensity PPML	
	Statistic	p	Statistic	p
LM lag	0.724	0.395	1.457	0.227
Robust LM lag	0.004	0.947	0.035	0.851
LM error	0.719	0.396	2.401	0.121
Robust LM error	0.000	0.994	0.979	0.322

Panel A gives the year of the most recent research and development expenditure observation for each estimation country. Panel B compares the reported cluster-robust standard errors with quasi-Poisson standard errors formed from the Pearson dispersion, $\hat{\phi}_P = 286$. The deviance-based dispersion for the same fit is $\hat{\phi}_D = 1.4$. The two disagree by more than two orders of magnitude because the Pearson statistic is dominated by a small number of systems with very large counts, so a constant variance-to-mean ratio does not describe these data and the quasi-Poisson interval is conservative rather than correct. Cluster-robust errors, which assume no variance function, remain the reported choice. Panel C reports Lagrange Multiplier tests for a spatial lag and a spatial error process; none rejects at the 5% level.

TABLE A8 | Participation models by publication period

Covariate	Published 2000–2019 Estimate	Published 2000–2019 95% CI	Published 2020–2026 Estimate	Published 2020–2026 95% CI
Log harvested area	0.750	[0.60, 0.90]	0.748	[0.64, 0.86]
R&D expenditure (% GDP)	0.568	[0.26, 0.88]	0.286	[0.03, 0.54]
Research-need index	0.634	[−1.88, 3.15]	0.455	[−1.34, 2.25]
Log GDP per capita	0.077	[−0.75, 0.90]	0.275	[−0.42, 0.97]
Tertiary enrolment	−0.765	[−2.21, 0.68]	0.420	[−0.52, 1.36]
Internet use	1.650	[−0.86, 4.16]	1.236	[−0.92, 3.40]
Log population	−0.030	[−0.23, 0.17]	0.070	[−0.16, 0.30]
Country-crop systems	1,799		1,799	
Systems carrying a study	225		402	

The outcome is whether a country-crop system carries a study published in the stated period; the covariate panel is unchanged, because the capacity and need indicators are single most-recent observations rather than annual series. The models test whether the associations are stable across the literature's recent expansion, not whether covariates changed. Standard errors are clustered by country. Note that 2026 is a partial publication year in the corpus.

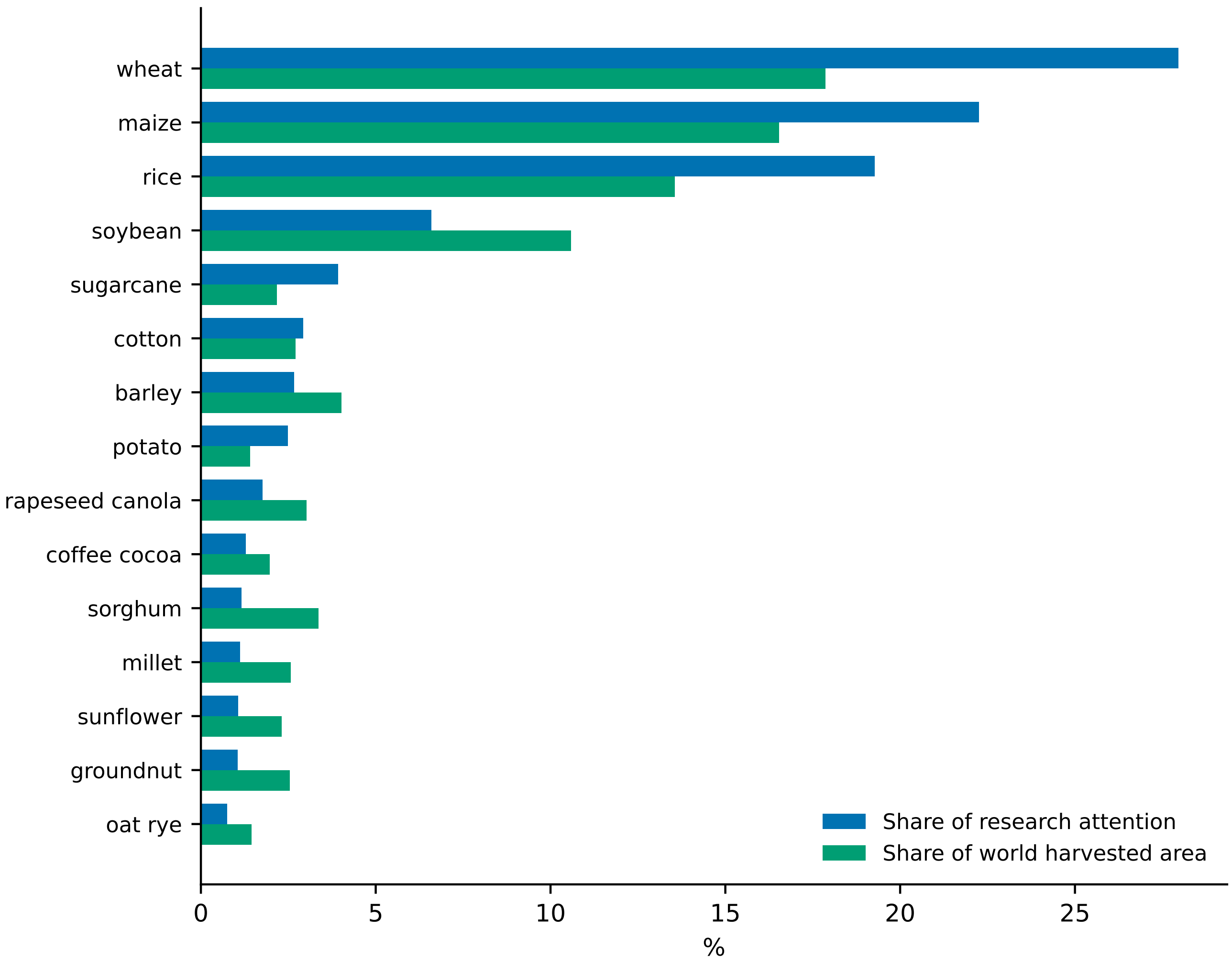

FIGURE A1 | Research attention and world harvested-area share, by crop, for the fifteen crops with the largest share of fractional research attention. Harvested area is the cross-crop benchmark because hectares, unlike tonnes, are comparable across crops.

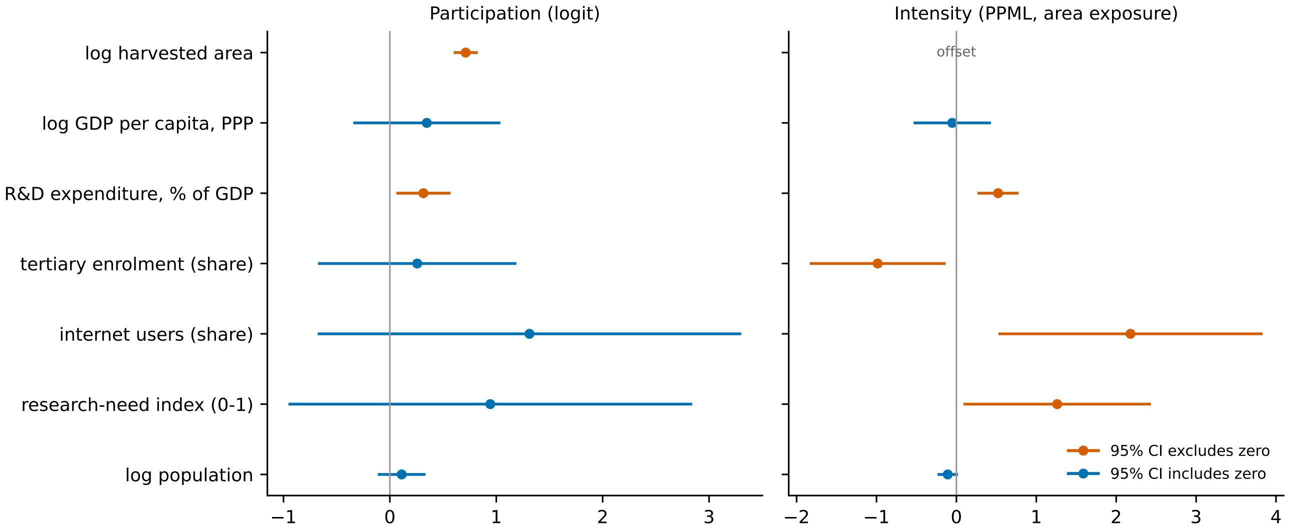

FIGURE A2 | Estimated coefficients for the participation and intensity models. Bars give 95% confidence intervals. The estimates are those reported in Table 7.

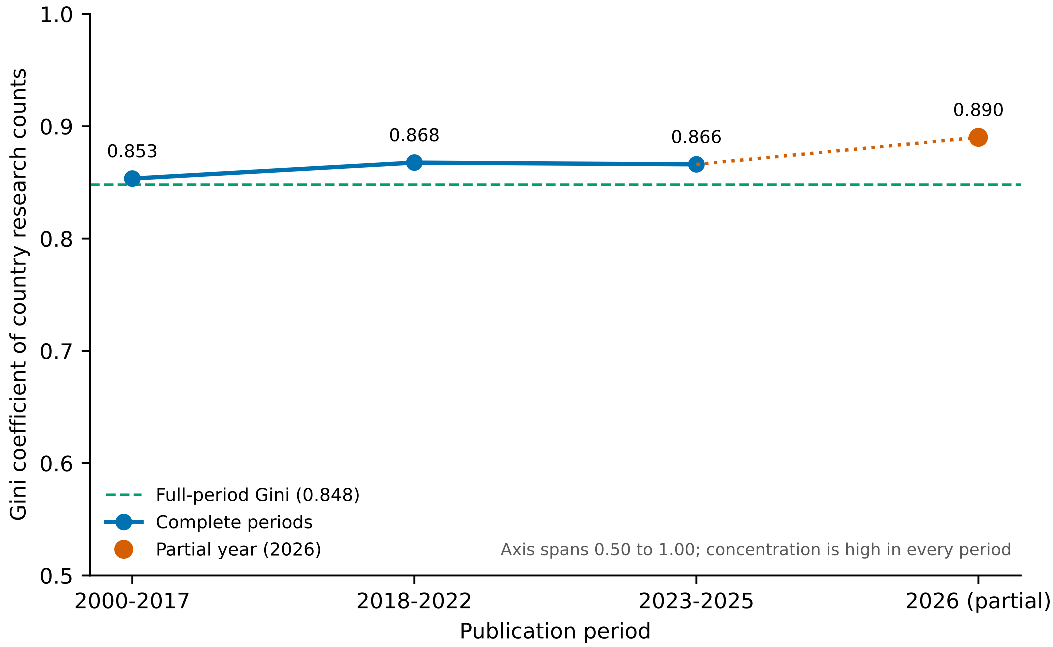

FIGURE A3 | Gini coefficient of country research counts by publication period, computed across all 195 countries in the boundary layer. The vertical axis spans 0.50 to 1.00 so that the small movement between periods is not visually inflated, and the dashed line marks the full-period value of 0.848. The 2026 point is drawn separately because that publication year is incomplete in the corpus.

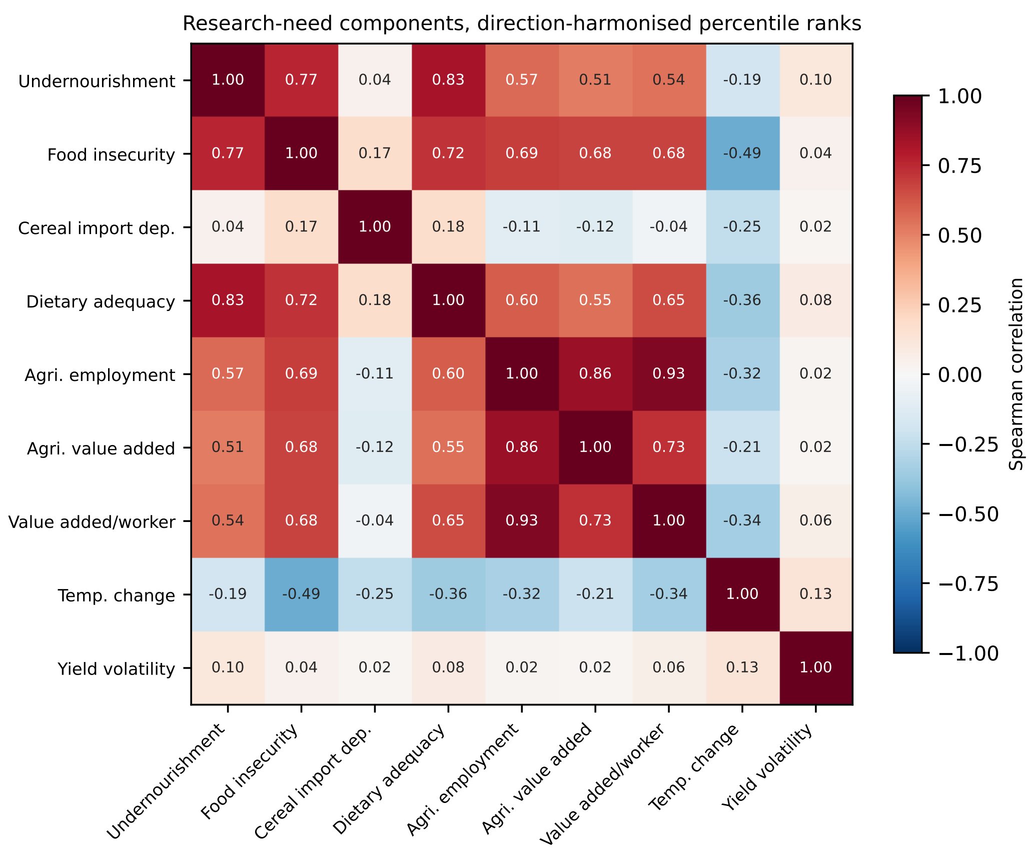

FIGURE A4 | Spearman correlations among the nine research-need components, after direction harmonisation and conversion to percentile ranks. The components are weakly related rather than interchangeable.